

19/12/25

Unit-1

Basics Of Operational Amplifier.

The electric device, active & passive components are fabricated on the single, chip integrated circuit (IC) which is a chip made of silicon material.

Advantage of Integrated circuits:

1. Smaller in size.
2. It is highly reliable.
3. Low cost.
4. Low power consumption.

5. Low Noise.

[The main factor for power consumption is speed]

Amplifier:

It strengthens the weaker signal.

op-amp:

We can implement mathematical

function along with amplifications.

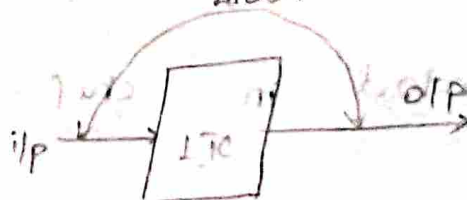
eg: it performs also operations like adding, integration, comparison etc...

LIC:

Relation btw input and output

of the particular IC is linear

that's why it is linear integrated circuits.



Classification of IC:

- Based on types of signals.

1. Analog IC - process only analog signal.

eg: Regulated IC 741

2. Digital IC - process only digital sig

eg: AND, OR.

3. Mixed signal IC.

eg: ADC, DAC, PLL.

- Based on level of integration.

1. SSI - small scale integration.

$\Rightarrow$ 1-10 no. of input gates
[first level of integration]

2. MSI - Medium scale integration

$\Rightarrow$ 10-100 no. of input gates can be integrated.

3. LSI - Large scale integration.

$\Rightarrow$ More than 1000 of inputs can

gates (or) transistors can be integrated.

4. VLSI - Very large scale integration.

$\Rightarrow$ millions of input gates can be integrated in chip.

- Based on fabrication technology.

$\Rightarrow$ Thin film.

$\Rightarrow$ Thick film.

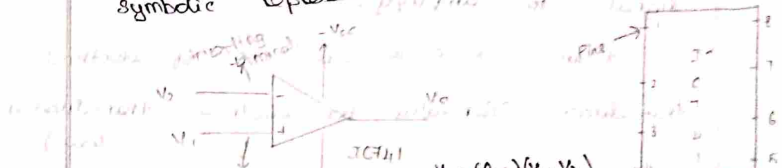
etc...

op-amp $\Rightarrow$ IC 741.

$\rightarrow$ indicates the name of the company which involves manufacturing process.

Internal blocks of operational amplifier:

Symbolic representation of IC.



$$V_0 = (A_{OL})(V_2 - V_1)$$

$A_{OL} \rightarrow$ open loop voltage gain of op-amp

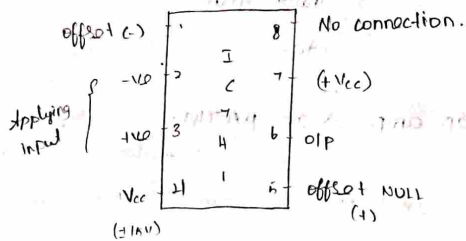
* If the IC which has pins on

both sides is called DIP [Dual In Line Packaging].

* If the IC which has pins on single side is called SIP [Single In line packaging].

* If the IC which has pins on all four sides of IC is called QFP [Quad Flat packaging]

BAP [Blue array packaging] - pins on the



supply voltage $\rightarrow \pm 12V$ to $\pm 15V$
opamp: It is used whenever the lowest signal to amplify.

assume V_1 & V_2 are o/p of electrical transducer (generally for electrical transducer is low)

$\Rightarrow$ 4 blocks are involved in op-amp.

* Differential amplifier.

[Dual input and dual ended o/p]

is called balance differential amplifier.

$\Rightarrow$ The phase relation b/w i/p & o/p of transistor is 180° .

If the input is given in two terminal it doesn't provide phase-shift.

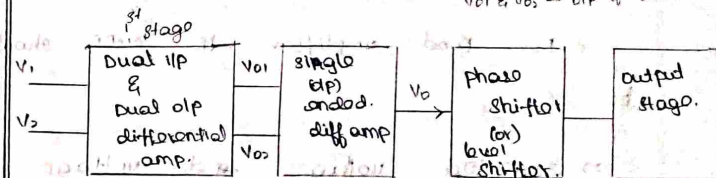
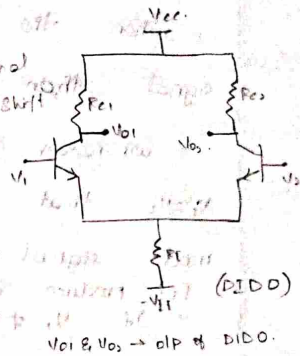
If the i/p is given in 1-ve terminal of op-amp it provide 180°

phase shift that's why is called inverting terminal.

$\Rightarrow$ It amplifies weaker signal by DDO with some phase shift. Then use DDO next.

Phase shifter provide proper 180° phase shift the last 4th

is output stage (it has to drain



characteristics:

The input impedance (R_i) of op-amp is correct (accurate).

① $\Rightarrow$ In ideal case, $R_i = \infty$ (In range 10^5 to $10^6 \Omega$ practically.)

dp impedance, $R_o = 0$

② $\Rightarrow$ In practical, $R_o < 100\Omega$.

③ $\Rightarrow$ BW = ∞ (ideal case) indicates high frequency.

It is range of frequency $\rightarrow$ BW. for which the circuit exhibit good performance.

④ $\Rightarrow$ (Common mode Rejection Ratio) CMRR = $\frac{A_d}{A_c}$ = differential mode gain / common mode gain.

* Gain is ratio of o/p to i/p.

* when the input is common mode signal then the gain $A_c = 0$.

* whenever V_1 & V_2 are same (common mode signal), the o/p is A_c ($A_c = 0$)

[it produce '0' amount of o/p]

* If $V_1 \neq V_2$, the gain circuit is A_d

* For good amplifier, the CMRR should be ∞ .

$\Rightarrow$ unwanted voltage or dc voltage existing in input of operational amplifier is known as offset voltage.

[presence of offset of V_{os} , output $\neq 0$] so.

By using offset (1 & 5 pin) we can avoid the offset voltage, to (cancel)

produce 'zero' o/p of the amplifier.

⑤ $\Rightarrow$ PSRR = ∞ [power supply rejection ratio]

The o/p of op-amp should not change even though some change in power supply. (It should not affect o/p)

in order to provide stable output.

It is defined as ratio of change in input offset voltage (ΔV_{os}) to change in supply voltage (ΔV_s).

$$PSRR = \frac{\Delta V_{os}}{\Delta V_s} = \text{PSRR [supply voltage rejection ratio]}$$

⑥. Zero Noise.

[No internal noise is produced in operational - amplifier].

Advantages of cascading the amplifier:

* Increase the overall gain of the system.

Internal blocks of op-amp.

1st stage. (DIDO op-amp)

* It ensures that input impedance is very low.

* amplify the signal.

2nd stage.

* Voltage gain $\uparrow$

output stage. (V_o) it's o/p is in usable form.

* To ensure that the o/p impedance is low.

* Supply large amount of current

to drive the output circuit.

[op-amp amplifier cannot produce more than ± 15 V.]

Q. If supply voltage is $\pm 10\text{ V}$, saturated voltage of op-amp is $\pm 12\text{ V}$.

If $V_{CC} = \pm 15V$ then saturated voltage of op-amp is $\pm 15V$.

characteristics:

- 1) $R_{in} = \infty$ (practical $10^5 - 10^6 \Omega$)
- 2) $R_o = 0$ ($< 100 \Omega$)
- 3) $BW = \infty$
- 4) $A_{OL} \neq \text{open loop gain} = \infty$ ($> 10^6$)
- 5) $CMRR = \text{Common Mode Rejection Ratio}$

(For good amplifier,
(or) ideal)

$$CMRR = \frac{A_{dm}}{A_{cm}} = \infty$$

$\Rightarrow$ DDO is also called symmetrical

differential amplifier.

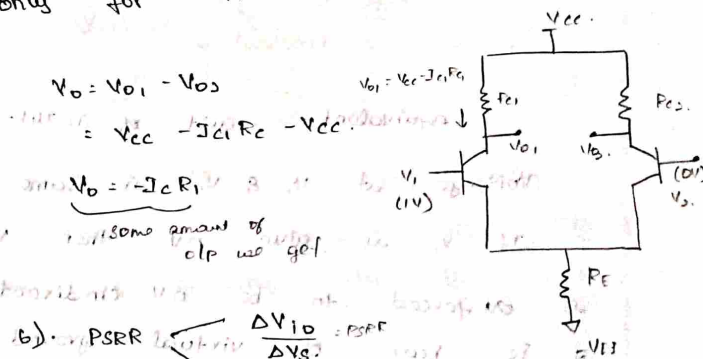
C-H's proportions are same like

Resistors, Transistors)

purpose of DDO:

- It reject all common Mode signal

The differential amplifier responds only for different inputs of amplifier.


$$b) \cdot \text{PSRR} \begin{cases} \frac{\Delta V_{IO}}{\Delta V_S} = \text{PSRR} \\ \frac{\Delta V_O}{\Delta V_S} = \text{PSRR} \end{cases}$$

PSRR & CMRR can be expressed in dB (or) normal.

7). zero noise.

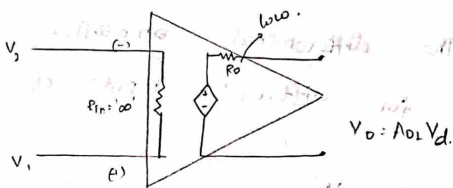
8). zero drift.

$\Rightarrow$ op-amp is not affected when temp in temperature.

2. a). zero i/p offset voltage. (in ideal case no i/p offset voltage).

10) slow Rate = '∞' (ideal) measured in terms of $\frac{V}{\mu s}$
 Max. Rate of change of output w.r.to time.

Virtual Ground: (V_G)



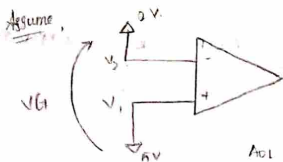
equivalent circuit of IC 741.

Voltage at V_1 & V_2 is same

If V_1 is given 0V then V_2

is forced to be 0V (indirectly)

It is bcoz of virtual ground of ideal op-amp.



output of op-amp. $V_0 = A_{01}(V_1 - V_2)$

$$= A_{01}(5 - 2)$$

$$V_0 = 3 \times 10^5$$

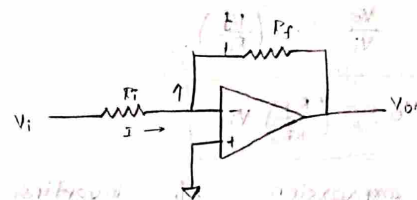
Since the o/p is more than 15V.

in this case the output of

op-amp is saturated output 15V.

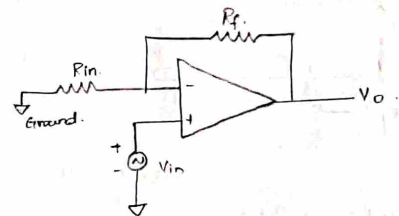
how quickly the output of op-amp responds to i/p (or) slow rate, measured by.

Inverting amplifier:

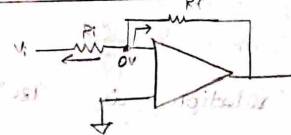


There is no current flow through input of op-amp due to $R_i = \infty$

Non-Inverting amplifier:



1. Inverting amplifier:



apply Nodal analysis.

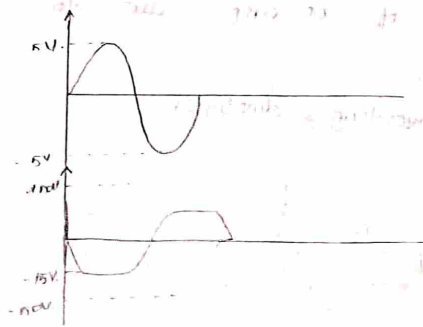
$$\frac{0 - V_i}{R_i} + \frac{0 - V_o}{R_f} = 0$$

$$\frac{-V_i}{R_i} = \frac{V_o}{R_f}$$

$$\frac{V_o}{V_i} = - \left(\frac{R_f}{R_i} \right)$$

$$V_o = - \left(\frac{R_f}{R_i} \right) V_i$$

output expression of inverting amplifier.



If, $R_f = 10k\Omega$, $R_i = 1k\Omega$.

$$V_o = - \left(\frac{10k}{1k} \right) V_i$$

$$= -10 \times 5 = -50V$$

⇒ phase relation is 180° phase shift.

$$A_{cl} = \frac{V_o}{V_i} = - \frac{R_f}{R_i}$$

A_{cl} → gain of closed loop.

Design an amplifier circuit to produce the gain of -5 .

If gain is $-ve$ then it is inverting amplifier.

Given, Gain $= -5$.

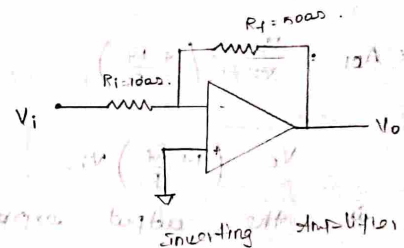
$$A_{cl} = - \frac{R_f}{R_i} = -5$$

$$R_f = 5 R_i$$

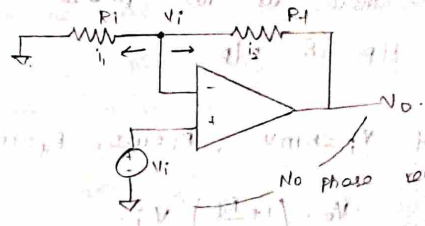
we choose, $R_i = 100\Omega$.

$$R_f = 5 \times 100\Omega$$

$$R_f = 500\Omega$$



2. Non-inverting amplifier.



By using Virtual Ground concept

Non-inverting terminal is V_i , so nodal

at inverting terminal is V_i

The current flows from higher potential to lower potential.

$$\frac{V_i - 0}{R_i} + \frac{V_i - V_o}{R_f} = 0.$$

$$\frac{V_i}{R_i} + \frac{V_i}{R_f} = \frac{V_o}{R_f}.$$

$$V_i \left[\frac{1}{R_i} + \frac{1}{R_f} \right] = \frac{V_o}{R_f}$$

$$\frac{V_o}{V_i} = \frac{R_i + R_f}{R_f} \times R_f.$$

$$\text{Gain} = A_{v1} = \frac{V_o}{V_i} = \left(1 + \frac{R_f}{R_i} \right)$$

$$V_o = \left(1 + \frac{R_f}{R_i} \right) V_i.$$

This is the output expression for Non-Inverting amplifier.

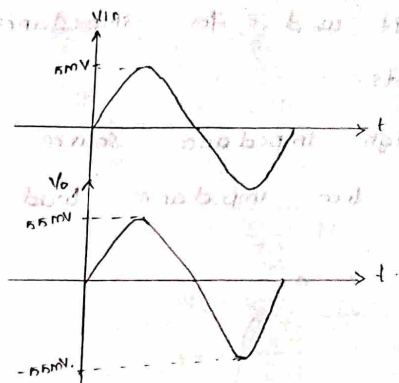
⇒ There is no phase relationship b/w i/p & o/p.

eg: if $V_i = 5 \text{ mV}$, $R_i = 10 \Omega$, $R_f = 100 \Omega$.

$$V_o = \left(1 + \frac{R_f}{R_i} \right) V_i.$$

$$= \left(1 + \frac{100}{10} \right) 5 \times 10^{-3}$$

$$V_o = 55 \text{ mV}.$$

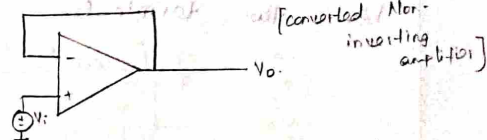


3. VOLTAGE FOLLOWER.

$$R_f = 0 \Omega.$$

$$R_i = \infty \Omega.$$

* The non-inverting amplifier is converted by assuming $R_f = 0 \Omega$, $R_i = \infty \Omega$



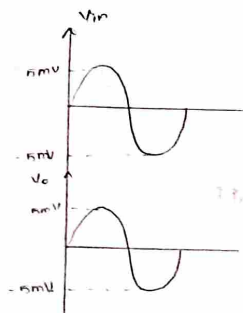
* The output simply follows the input signal. That's why it is called voltage follower.

$$\text{Gain} = \frac{V_o}{V_i} = 1 \quad [\because V_o = V_i]$$

* also known as unity gain amplifier circuit.

It used for impedance matching circuits.

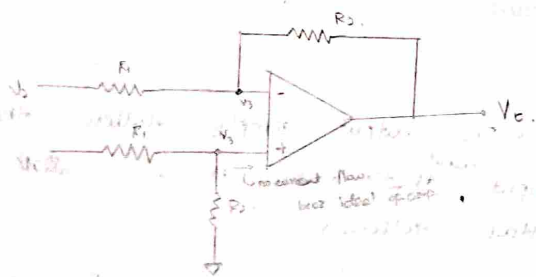
High impedance source is connected with low impedance load $\Rightarrow$ impedance matching.



i/f & o/p waveforms are same.

DIFFERENTIAL AMPLIFIER / DIFFERENCE AMPLIFIER.

It will amplify the difference b/w two terminals.



Use virtual ground concept.

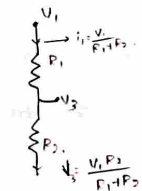
$$\frac{V_3 - V_1}{R_1} + \frac{V_3 - 0}{R_2} = 0$$

$$\frac{V_3}{R_1} + \frac{V_3}{R_2} = \frac{V_1}{R_1}$$

$$V_3 \left[\frac{1}{R_1} + \frac{1}{R_2} \right] = \frac{V_1}{R_1}$$

$$V_3 = \frac{V_1}{\frac{R_1 + R_2}{R_1 R_2}}$$

$$V_3 = \frac{V_1 R_2}{R_1 + R_2} \rightarrow \text{①}$$



R_1 & R_2 series.

Apply nodal analysis w.r.t (-ve)

terminal of op-amp :

$$\frac{V_3 - V_2}{R_1} + \frac{V_3 - V_0}{R_2} = 0$$

$$V_3 \left[\frac{1}{R_1} + \frac{1}{R_2} \right] - \frac{V_2}{R_1} = \frac{V_0}{R_2}$$

$$V_3 \left[\frac{R_1 + R_2}{R_1 R_2} \right] - \frac{V_2}{R_1} = \frac{V_0}{R_2} \rightarrow \text{②}$$

Sub V_3 ,

$$\frac{V_1 R_2}{R_1 + R_2} \left[\frac{R_1 + R_2}{R_1 R_2} \right] - \frac{V_2}{R_1} = \frac{V_0}{R_2}$$

$$\frac{V_1}{R_1} - \frac{V_2}{R_1} = \frac{V_0}{R_2}$$

$$\frac{1}{R_1} [V_1 - V_2] = \frac{V_0}{R_2}$$

$$V_o = \frac{R_2}{R_1} [V_1 - V_2]$$

* The output expression is amplified version of difference btw V_1 & V_2 .

eg: $V_1 = 3\text{mV}$, $V_2 = 2\text{mV}$.

$$R_1 = 10\text{k}\Omega, R_2 = 1000\text{k}\Omega$$

$$V_o = \frac{R_2}{R_1} (V_1 - V_2)$$

$$= \frac{1000\text{k}}{10\text{k}} (3\text{m} - 2\text{m})$$

$$= 100 (1\text{m}) \text{ V}$$

$$V_o = 0.1 \text{ V}$$

* Instrumentation amplifier means amplifies the weak (or) low voltage signal.

* It is used in instrumentation amplifier.

(adv of difference amplifier)

* High input impedance & low output impedance.

For amplifier circuit, the transistor should in the linear mode for that biasing method is used.

$$V_d = (V_1 - V_2) \text{ difference btw two terminal.}$$

$$V_{cm} = \frac{V_1 + V_2}{2} \text{ average of two terminal.}$$

$$V_o = A_1 V_1 + A_2 V_2 \rightarrow \textcircled{1}$$

consider $A_1 = \frac{V_o}{V_1}$ at $V_2 = 0$

$$A_2 = \frac{V_o}{V_2} \text{ at } V_1 = 0$$

$$\left. \begin{aligned} V_1 &= V_{cm} + \frac{V_D}{2} \\ V_2 &= V_{cm} - \frac{V_D}{2} \end{aligned} \right\} \rightarrow \textcircled{2}$$

sub eq $\textcircled{2}$ in $\textcircled{1}$

$$V_o = A_1 \left[V_{cm} + \frac{V_D}{2} \right] + A_2 \left[V_{cm} - \frac{V_D}{2} \right]$$

$$V_o = V_{cm} [A_1 + A_2] + \frac{V_D}{2} [A_1 - A_2]$$

$$V_o = A_{cm} V_{cm} + A_{dm} V_D \rightarrow \textcircled{3}$$

compare equ $\textcircled{1}$ & $\textcircled{3}$

$$A_{cm} = A_1 + A_2$$

$$A_{dm} = \frac{1}{2} (A_1 - A_2)$$

$$\delta = \text{CMRR} = \left| \frac{A_{DM}}{A_{CM}} \right| \rightarrow \text{ideal op-amp} = \infty$$

$$\rightarrow \text{Practical op-amp} \approx 70 \text{ dB}$$

DC CHARACTERISTICS OF OP-AMP:

$\Rightarrow$ whenever the op-amp is non-ideal.

it defined as DC & AC characteristics.

$\Rightarrow$ Non-ideal op-amp $\rightarrow$ DC characteristics.

$\Rightarrow$ 4 DC characteristics.

- 1). Input Bias current.
- 2). Input offset current.
- 3). Input offset voltage.
- 4). Thermal drift.

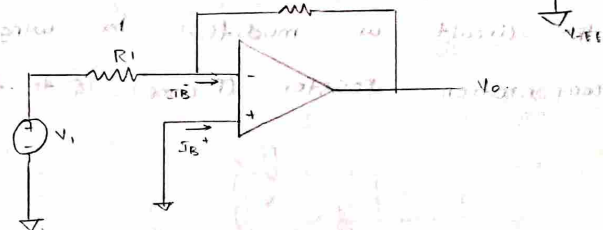
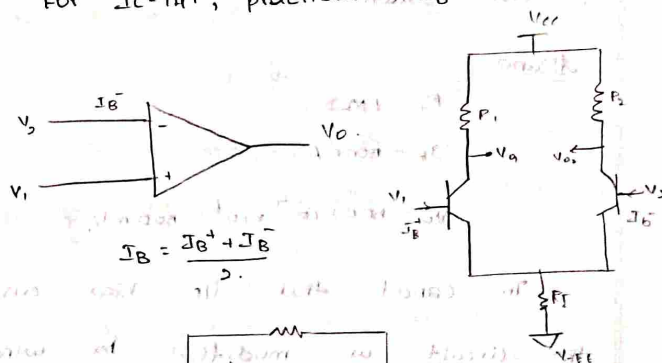
① Input Bias Current.

I_B^- $\rightarrow$ current enters to the inverting terminal.

There must of small amount of current flow through the base of the transistor. There must be some imbalance so in practical I_B^+ , I_B^- it is not same

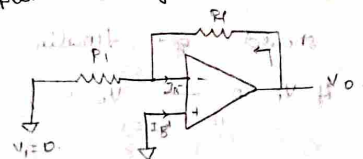
$I_B \rightarrow$ average of current flowing through two terminal of op-amp is known as input bias current.

For IC-741, practical $I_B = 500 \text{ nA}$.



($V_1 = 0 \rightarrow V_0 = 0$).

Due to some bias current the output voltage is not zero.



$$V_0 = -I_B^- R_f$$

Even though the input is zero the output has some value instead of getting 'zero' it is bcoz due to i/p bias current.

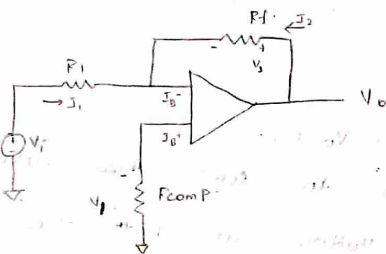
Assume:

$$R_f = 1 \text{ M}\Omega$$

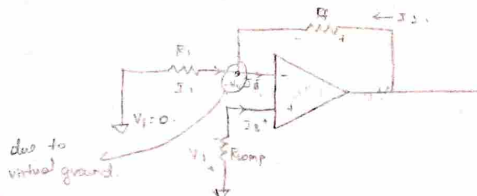
$$I_B = 500 \text{ nA}$$

$$V_o = 500 \times 10^{-9} \times 10^6 = 500 \text{ mV}$$

To cancel this i/p bias current, the circuit is modified, by using compensation Resistor (R_{comp}). & to find R_{comp} .



[the circuit should get finally if $V_i = 0 \rightarrow V_o = 0$]



Apply KVL,

$$V_o - V_2 - (-V_1) = 0$$

$$V_o = V_2 - V_1 \rightarrow (1)$$

to achieve $[V_i = 0 \rightarrow V_o = 0]$.

$$\therefore V_1 = V_2 \rightarrow (2)$$

The voltage across R_{comp} & voltage across feedback resistor should be equal.

The compensation circuit is valid only by considering that $I_B^- = I_B^+$ [circuits work only (condition is same bias current flows through two terminal)]

$$I_B^+ = \frac{V_1}{R_{comp}} \rightarrow (3)$$

$$I_1 = \frac{V_i - (-V_1)}{R_f} = \frac{0 + V_1}{R_f}$$

$$I_1 = \frac{V_1}{R_f} \rightarrow (4)$$

$$\text{by nodal, } I_B^- = I_1 + I_2$$

$$I_B^- = \frac{V_1}{R_f} + \frac{V_2}{R_f} \rightarrow (5)$$

equating equation (3) & (5).

$$\frac{V_1}{R_{comp}} = \frac{V_1}{R_1} + \frac{V_2}{R_f}$$

$$[\because V_1 = V_2]$$

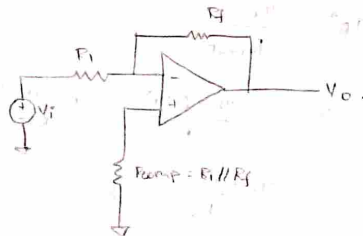
$$\frac{V_1}{R_{comp}} = \frac{V_1}{R_1} + \frac{V_1}{R_f}$$

$$\frac{V_1}{R_{comp}} = V_1 \left[\frac{1}{R_1} + \frac{1}{R_f} \right]$$

$$\frac{1}{R_{comp}} = \frac{1}{R_1} + \frac{1}{R_f}$$

$$\therefore R_{comp} = \frac{R_1 R_f}{R_1 + R_f} = R_1 \parallel R_f$$

to compensate the effect due to i/p bias current, the ($R_{comp} = R_1 \parallel R_f$) is connected in non-inverting terminal.



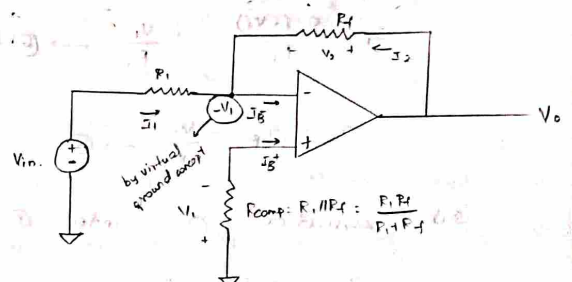
②. Input offset current. (I_{os})

It is difference between two terminal current of op-amp.

Generally,

$$|I_{os}| = I_B^+ - I_B^- = I_B^- - I_B^+$$

In practical value of $I_{os} = 200 \text{ nA}$



If two terminal current is not same then we use this.

$$[KVL] \quad V_o - V_2 - (-V_1) = 0$$

$$V_o - V_2 + V_1 = 0$$

$$V_o = V_2 - V_1 \rightarrow (1)$$

$$V_2 = I_2 R_f \rightarrow (2) \quad \frac{R_f}{-V_2} \leftarrow I_2$$

$$V_1 = I_B^+ R_{comp} \rightarrow (3)$$

I_B^- & I_B^+ are not equal

if so what is equation we find.

write KCL equation w.r.t (-ve) terminal of op-amp.

$$I_1 + I_2 = I_B^-$$

$$I_2 = I_B^- - I_1$$

Assume, $V_i = 0$.

$$I_1 = \frac{0 - (-V_i)}{R_1} = \frac{V_i}{R_1} \rightarrow (4)$$

$$\therefore I_2 = I_B^- - \frac{V_i}{R_1} \rightarrow (5)$$

Sub equation (3) into (5).

$$I_2 = I_B^- - \frac{I_B^+ R_{comp}}{R_1} \rightarrow (6)$$

Apply equation (5), (3), (6) into eqn (1).

$$V_o = I_2 R_f - I_B^+ R_{comp}$$

$$V_o = \left[I_B^- - \frac{I_B^+ R_{comp}}{R_1} \right] R_f - I_B^+ R_{comp}$$

$$V_o = I_B^- R_f - I_B^+ R_{comp} \left[\frac{R_f}{R_1} + 1 \right]$$

$$\therefore V_o = I_B^- R_f - I_B^+ R_{comp} \left[\frac{R_1 + R_f}{R_1} \right] \cdot \frac{R_f}{R_f}$$

$$V_o = I_B^- R_f - I_B^+ R_{comp} \left[\frac{R_1 + R_f}{R_1 R_f} \right] R_f$$

$$V_o = I_B^- R_f - I_B^+ R_f$$

$$V_o = (I_B^- - I_B^+) R_f$$

$$V_o = |I_{os}| R_f$$

For JCT41, $|I_{os}| = 200 \text{ nA}$ assume, $R_f = 1 \text{ M}\Omega$.

$$V_o = 200 \times 10^{-9} \times 1 \times 10^6$$

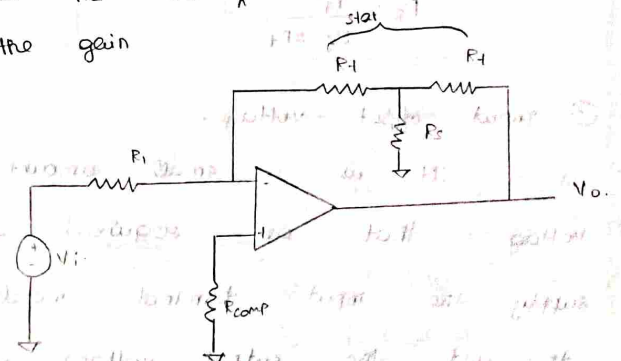
$$= 200 \times 10^{-3}$$

$$V_o = 200 \text{ mV}$$

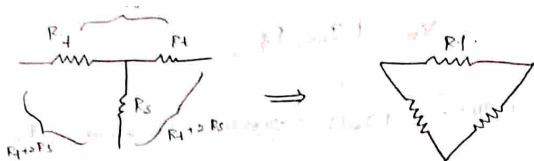
When no i/p applied but V_o is getting 200 mV because of i/p offset current.

if $|I_{os}| \uparrow$ so, $V_o \uparrow$ so we need to cancel this effect.

We need to choose proper R_f value to reduce this effect. at the same time it should not affect the gain.



modified circuit to cancel i/p offset current at the same time not affect the gain.



[T-network]

$$R_{AB} = \frac{R_A + R_B + R_C}{R_C}$$

$$R_f = R_1 + R_2 + \frac{R_1 R_2}{R_3}$$

$$R_f = \frac{2R_1 R_3 + R_1^2}{R_3}$$

choose, $R_1 \ll R_3$

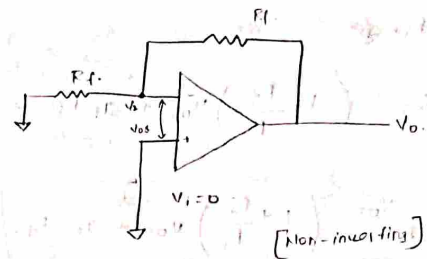
$$2R_1 + \frac{R_1^2}{R_3} = R_f$$

$$\frac{1}{R_3} = \frac{R_f - 2R_1}{R_1^2}$$

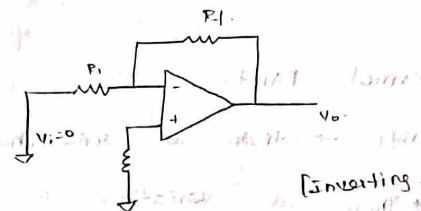
$$R_3 = \frac{R_1^2}{R_f - 2R_1}$$

3. Input offset voltage.

It is a small amount of dc voltage that are required to supply the input terminal in order to get the output voltage is zero (when V_1 & $V_2 = 0$).



[Non-inverting]



[Inverting]

Voltage divided rule.

$$V_2 = \frac{V_o}{R_1 + R_f} \cdot R_1$$

$$V_{os} = V_1 - V_2$$

$\therefore V_1 = 0$

$$|V_{os}| = V_2$$

$$V_{os} = \frac{V_o}{R_1 + R_f} \cdot R_1$$

$$\therefore V_o = \frac{R_1 + R_f}{R_1} \cdot V_{os}$$

output voltage affected by for non-ideal op-amp.

$$I_B \rightarrow V_o = I_B R_f$$

$$I_{os} \rightarrow V_o = I_{os} R_f$$

$$V_{os} \rightarrow V_o = \left(1 + \frac{R_f}{R_1}\right) V_{os}$$

Total changes.

[When
Ramp is
not known]

$$V_{OT} = \left(1 + \frac{R_f}{R_i}\right) V_{OS} + \underbrace{I_B R_f + I_{AS} R_f}_{\text{compensated by using Ramp neglected.}}$$

[R_i is
known
& designed]

$$V_{OT} = \left(1 + \frac{R_f}{R_i}\right) V_{OS} + \underbrace{I_{AS} R_f}_{\text{V_p offset current affects strongly.}}$$

④ Thermal Drift.

drift → there is some changes.

* There is variations in 3 other
DC characteristics depends on temperature.

Representation (a) expressed, terms.

⇒ it drift happens on.

$$\left. \begin{array}{l} \text{IIP bias current} \\ \text{IIP offset current} \end{array} \right\} \rightarrow \text{nA/}^\circ\text{C}$$

⇒ it drift happens on.

$$\text{IIP offset voltage} \rightarrow \text{mV/}^\circ\text{C}$$

we can prevent the thermal drift

* by proper PCB design.

* using cooling circuits.

①

Design an inverting amplifier to compensate the effect due to IIP offset current using 12TH op-amp to get a gain of -10 and IIP impedance of 10 MΩ.

Sol:

$$A_{CL} = \frac{-R_f}{R_i} = -10.$$

$$\text{Given, } R_i = 10 \text{ M}\Omega.$$

$$R_f = 10 \times 10 \text{ M}\Omega.$$

$$R_f = 100 \text{ M}\Omega.$$

$$R_f \ll R_{f/2}$$

$$R_f \ll \frac{100}{2} \text{ M}\Omega.$$

$$R_f \ll 50 \text{ M}\Omega.$$

$$\text{choose } R_f = 45 \text{ M}\Omega.$$

$$R_3 = \frac{R_f^2}{R_f - 2R_f}$$

$$= \frac{(45 \times 10^6)^2}{100 \times 10^6 - 2 \times 45 \times 10^6}$$

$$= \frac{2025 \times 10^{12}}{10^6 (100 - 90)}$$

$$= \frac{2025}{10} \times 10^6$$

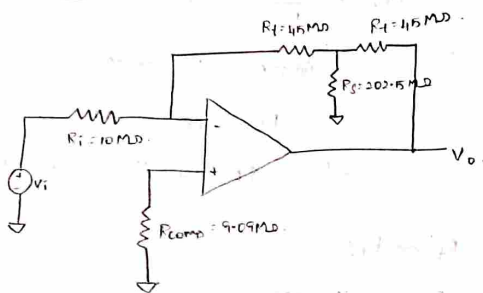
$$R_3 = 202.5 \text{ M}\Omega$$

$$R_{comp} = R_i || R_f$$

$$= \frac{R_i R_f}{R_i + R_f} = \frac{10 \times 10^6 \times 100 \times 10^6}{110 \times 10^6}$$

$$= \frac{1000}{110} \times 10^6$$

$$R_{comp} = 9.09 \text{ MD.}$$



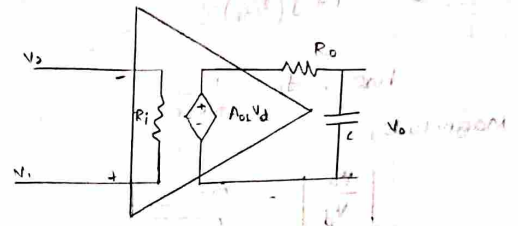
AC CHARACTERISTICS OF OPERATIONAL AMPLIFIER:

- 1). Frequency Response.
- 2). Slew Rate.
- 3). Stability.

[What about the output of system when the frequency changes - frequency Response.]

①. Frequency Response:

$$V_o = \frac{A_{ol} (V_i - V_o)}{A_{ol} V_d}$$



C → due to parasitic capacitor existing in the internal op-amp.

R_o → output Resistance (ideal $R_o = 0$)

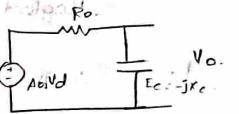
R_i → input Resistance (for ideal $R_i = \infty$)

A_{ol} → open loop gain.

V_d → difference b/w 2 terminals voltage.

By voltage divided,

$$V_o = \left[\frac{A_{ol} V_d}{R_o - jX_c} \right] (-jX_c)$$



$$[\because X_c = \frac{1}{\omega C}]$$

$$V_o = \frac{A_{ol} V_d}{-jX_c [1 + j \frac{R_o}{X_c}]} (-jX_c)$$

$$V_o = \frac{A_{ol} V_d}{1 + j \omega R_o C}$$

$$V_o = \frac{A_{ol} V_d}{1 + j \omega R_o C}$$

$$\frac{V_o}{V_d} = \frac{A_{ol}}{1 + j \omega R_o C}$$

$$\frac{V_o}{V_d} = \frac{A_{ol}}{1 + j \omega R_o C}$$

$$\frac{V_o}{V_d} = \frac{A_{OL}}{1 + j(f/f_L)}$$

$f/f_L \rightarrow$ frequency due to op resistance & ip resistance

here, $f_L = \frac{1}{2\pi R_o C}$

Magnitude,

$$\left| \frac{V_o}{V_d} \right| = \frac{A_{OL}}{\sqrt{1 + (f/f_L)^2}}$$

$f_L \rightarrow$ can't change
 $f \rightarrow$ can vary.

Cases:

i) $f = f_L \Rightarrow \left| \frac{V_o}{V_d} \right| = \frac{A_{OL}}{\sqrt{2}}$

ii) $f \ll f_L \Rightarrow f/f_L \ll 1$

* Neglect the term f/f_L compare to

i. $\therefore$ The gain is constant. (same as A_{OL})

$$\left| \frac{V_o}{V_d} \right| = A_{OL}$$

iii) $f \gg f_L \Rightarrow f/f_L \gg 1$

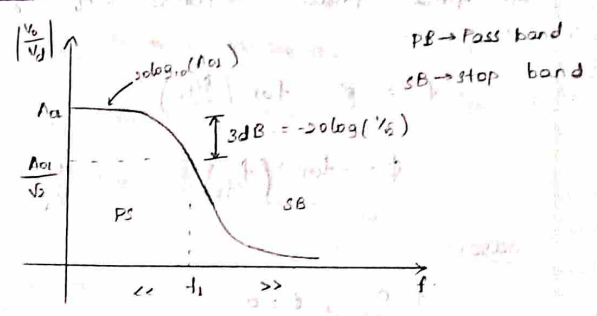
* Neglect 1 compared to the term f/f_L

$$\left| \frac{V_o}{V_d} \right| = \frac{A_{OL}}{\sqrt{(f/f_L)^2}}$$

$$\left| \frac{V_o}{V_d} \right| = \frac{A_{OL} f_L}{f}$$

f increases then the gain $\frac{V_o}{V_d}$ decreases.

Magnitude Response:



f_L $\Rightarrow$ 3db frequency - frequency at which gain is reduced to.

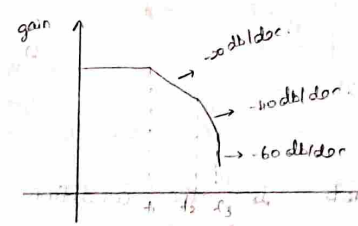
70.7% of its Max gain
0.707 Max gain.

PE $\rightarrow$ Range of frequency which allow all the frequency (0 to 10 kHz)

SB $\rightarrow$ from (10 kHz) it attenuates.

* for order of system is 2nd then -20 dB/dec.

[assume n no. of op-amp is cascaded.]



to find phase response of system.

$\phi = 0^\circ - \tan^{-1}\left(\frac{f}{f_1}\right)$
 → No val part in Numerator.

$\phi = -\tan^{-1}\left(\frac{f}{f_1}\right)$

$\frac{V_o}{V_d} = \frac{A_{OL}}{1+j(f/f_1)}$

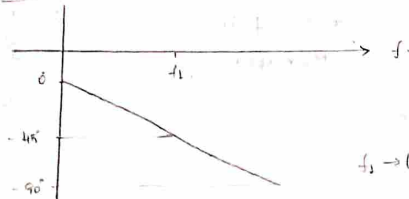
cases:

$f = 0, \phi = 0^\circ$

$f = f_1, \phi = -45^\circ$

$f = \infty, \phi = -90^\circ$

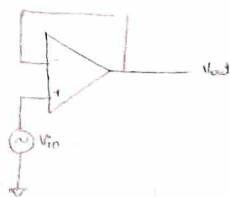
Phase Response:



$f_1 \rightarrow$ lies b/w 0° to 45°

$\Rightarrow$ max phase shift

$\approx -45^\circ$ for 1 RC circuit.



$V_{in} = V_{out}$

slow rate is the condition for which we get distortion ^{less} as output for input applied.

Max rate of change output w.r. to time for the applied input

It is measured in $V/\mu s$ unit.

$SR = \left. \frac{dV_o(t)}{dt} \right|_{\max}$

$\Rightarrow$ when capacitor is not there no sudden change in value (n) fall, it changes gradually.

to find slew rate:

$I = C \frac{dV_{out}}{dt}$

$\frac{dQ}{dt} = C \frac{dV_{out}}{dt}$

$I = C \frac{dV_{out}}{dt}$



It is not same as V_o due to presence of capacitance affect inside the op-amp.
 $\Rightarrow$ for ideal case 'SR' must be ∞

$SR \Rightarrow \frac{dV_{out}}{dt} = \frac{I}{C}$

for voltage follower circuit.

$V_{out} = V_{in} = V_m \sin \omega t$

$\frac{dV_{out}}{dt} = V_m \cos \omega t (\omega) = V_m \omega \cos \omega t$

$\left. \frac{dV_{out}}{dt} \right|_{\max} = V_m \omega = 2\pi V_m f_{\max} = SR$

$\therefore SR = \frac{2\pi V_m f_{\max}}{10^6} \text{ V}/\mu s$

Slow Rate:

* It depends on max frequency of input.

* It depends on maximum amplitude of input signal (V_m).

f_{max} - Maximum frequency of the input in order to get the distortion less output.

$\Rightarrow$ For JCT41, the slew rate is $0.5 \text{ V}/\mu\text{sec}$
 [SR should express only in $\text{V}/\mu\text{sec}$]

Q. A JCT41 op-amp is used as an inverting amplifier with gain of 50. The voltage gain vs frequency curve of JCT41 is flat upto 20 kHz , the slew rate of JCT41 is $0.5 \text{ V}/\mu\text{sec}$ what maximum peak to peak input signal can be applied without distorting the output.

Sol:

Given, $f_{max} = 20 \text{ kHz}$.

Gain = 50.

SR = $0.5 \text{ V}/\mu\text{sec}$.

$$SR = \frac{2\pi V_m f_{max}}{10^6} \text{ V}/\mu\text{s}.$$

$$0.5 = \frac{2\pi \times V_m \times 20 \times 10^3}{10^6}$$

$$\frac{0.5 \times 10^3}{2\pi \times 20} = V_m$$

$$V_m = \frac{500}{40\pi}$$

$$V_m = \frac{500}{125.6637}$$

$$V_m = 3.9788 \text{ V}$$

$$V_{p-p} = 2V_m = 2 \times 3.9788 \text{ V}.$$

$$V_{p-p} = 7.9577 \text{ V}.$$

$$A = \frac{V_o}{V_{in}}$$

($V_{in} = V_m$)

$$V_o = 50 \times 3.9788$$

$$V_o = 198.9 \text{ V}.$$

$$V_o = 50 \times 7.96$$

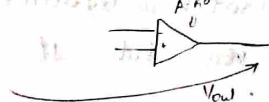
($V_{in} = V_{p-p}$)

$$V_o = 398 \text{ V}.$$

on $\text{V}/\mu\text{s} \leftarrow SR$

$$V_m = 3.9788 \text{ V}$$

$$V_{p-p} = 7.9577 \text{ V}$$



for this what V_{in} ?

$$V_o = A V_{in}$$

$$V_{in} = \frac{7.96}{50}$$

$$V_{in} = 0.159 \text{ V}$$

$$V_{in} = 159 \text{ mV}.$$

The i/p signal of 159 mV can be applied without distorting the output.

- ③. A non-inverting amplifier with V_o is null at 25°C . What will happen to the output voltage if temperature rises to 50°C for an offset voltage drift of $0.15\text{ mV}/^\circ\text{C}$.

Sol:

Given, $A=100$.

offset voltage drift $0.15\text{ mV}/^\circ\text{C}$.

$$V_o = A (50-25) \times 0.15\text{ mV}$$

$$V_o = 100 \times 25 \times 0.15 \times 10^{-3}$$

$$V_o = 375\text{ mV}$$

- ④. If the bias current in the JFET op-amp is 19 pA & the internal frequency compensation capacitor is 30 pF then find the slew rate of op-amp.

$$S.R. = \frac{I}{C}$$

$$= \frac{19 \times 10^{-6}}{30 \times 10^{-12}}$$

$$= \frac{0.633}{10^{-6}}$$

$$S.R. = 0.633\text{ V}/\mu\text{s}$$

Rise time for op-amp:

$$t_r = \frac{0.35}{\text{BW}}$$

time taken to output to reach its 10% to 90% of its maximum value.

- ③. stability:

* open loop circuit op-amp is not used to design amplifier circuit (not advisable)

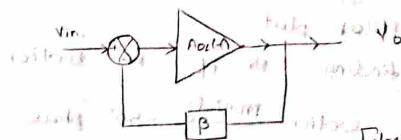
2nd

* due to $A_{OL} = \infty$,

when we apply $< 1\text{ mV}$, the output is amplified give o/p.

But we apply $> 1\text{ mV}$, the output is amplified ∞ due to gain is ∞ , but in op-amp maximum is supply voltage. so it is saturated, that's why is not preferable.

In comparator circuit we can use open loop op-amp circuits.



(closed loop negative feedback)

gain expression of closed loop operational amplifier.

$$\frac{V_o}{V_{in}} = \frac{A_{OL}(f)}{1 + A_{OL}(f)\beta}$$

The denominator of polynomial is characteristics eqn.

$$1 + A_{OL}(f)\beta = 0$$

$$A_{OL}(f)\beta = -1$$

If $A_{OL}(f)\beta$ is -1 , it becomes oscillatory, so we take magnitude.

$$|A_{OL}(f)\beta| = 1$$

* The order of system is $\uparrow$ as when considering more no. of capacitor effect

* If system is higher order system to find stability we can use.

1. Bode plot.

2. Nyquist plot.

3. Polar plot.

4. finding no. of RC section.

1 RC section $\xrightarrow{\text{provide}}$ -90° phase shift.

For multiple no. of RC $\xrightarrow{\text{provides}}$ multiple no. of phase shift.

(*) $\Rightarrow$ lower order system is always stable when compared to higher order system.

* stable when Bounded input & Bounded output.

* when it is linear circuit is stable.

This is attained by.

$\Rightarrow$ cancel out (or) reduce the RC section.

$\Rightarrow$ use frequency compensation.

when multiple corner frequency $\uparrow$ it leads to unstable system.

Frequency Compensation.

low (or) low

$\Rightarrow$ This reduces the no. of corner frequency exist in circuit.

1. Internal

Frequency compensation.

$\Rightarrow$ building capacitor inside the IC (built in)

[when 10-30 pF]

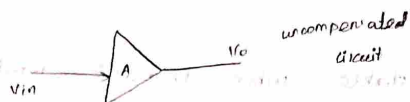
2. External frequency Compensation.

we need to combine the op-amp with additional circuit [containing capacitor].

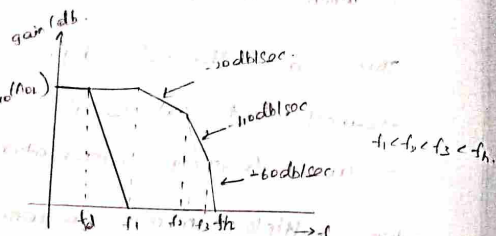
dominant pole.

Pole-zero compensation.

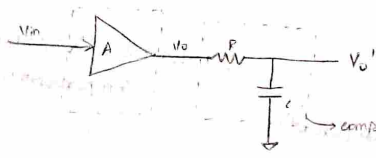
dominant-pole compensation



assume 3 corner frequencies



In order to compensate corner frequencies additional circuits are added.



$$\frac{V_o'}{V_{in}} = A \cdot A_1$$

for capacitor C , $X_C = \frac{1}{j\omega C} = -jX_C$;
(impedance) $X_C = 1/\omega C$.

$$V_o' = \frac{V_o (-jX_C)}{R + jX_C} = \frac{-jX_C \cdot V_o}{-jX_C [1 + jRX_C]}$$

$$A_1 = \frac{V_o'}{V_o} = \frac{1}{1 + jRX_C}$$

$$\frac{V_o'}{V_o} = \frac{1}{1 + j(f/f_d)}$$

here, $f_d = \frac{1}{2\pi RC}$; $f_d < f_1 < f_2 < f_3 < f_h$.

$f_d \rightarrow$ dominant frequency.

$\therefore$ choose proper value of R & C so that f_d is less than f_1, f_2, f_3

$$A = \frac{A_{OL}}{[1 + j(f/f_1)][1 + j(f/f_2)][1 + j(f/f_3)]}$$

$A_1 \rightarrow$ gain of compensated circuit (R & C)

$$A_{\text{gain}} = A \cdot A_1 = \frac{A_{OL}}{[1 + j(f/f_1)][1 + j(f/f_2)][1 + j(f/f_3)][1 + j(f/f_d)]}$$

(v) $\Rightarrow f_d$ must be less than f_1, f_2, f_3 .

$\Rightarrow f_d$ depends on R & C .

Drawback:

$\Rightarrow$ It is not suitable for application

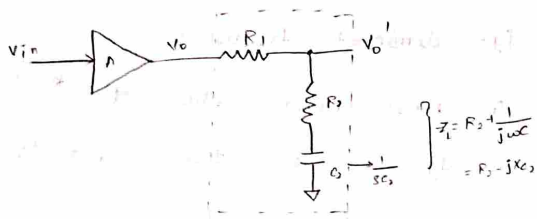
when BW is more.

$\Rightarrow$ It has less bandwidth.

To overcome this we use pole-zero technique.

How we include poles & zero to improve the circuit, that's why is called as dominant pole.

Pole-zero compensation:



How we include zero as well as pole, by which some improvement is done.

$$\frac{V_o'}{V_{in}} = A \cdot A_1$$

$$V_o' = \frac{V_o (R_2 - jX_{C_2})}{R_1 + R_2 - jX_{C_2}}$$

$$\text{here, } Z_1 = R_2 + \frac{1}{sC_2}$$

$$Z_1 = \frac{1 + sR_2C_2}{sC_2} \rightarrow \textcircled{1}$$

$V_o' \rightarrow$ voltage across Z_1

$$V_{o'} = \left(\frac{V_o}{R_1 + Z_1} \right) \cdot Z_1 \rightarrow \textcircled{2}$$

Sub ① in ②

$$V_o' = \frac{V_o \left(\frac{1 + sR_2C_2}{sC_2} \right)}{R_1 + \left(\frac{1 + sR_2C_2}{sC_2} \right)}$$

$$\frac{V_o'}{V_o} = A_1 = \frac{1 + sR_2C_2}{1 + sR_1C_2 + sR_2C_2}$$

$$A_1 = \frac{1 + sR_2C_2}{1 + sC_2(R_1 + R_2)}$$

$$A_1 = \frac{1 + j\omega R_2C_2}{1 + j\omega C_2(R_1 + R_2)}$$

$$A_1 = \frac{1 + j2\pi f R_2C_2}{1 + j2\pi f C_2(R_1 + R_2)}$$

$$A_1 = \frac{1 + j(f/f_1)}{1 + j(f/f_p)}$$

here,

$f_p \rightarrow$ frequency due to insertion of pole.

$$f_p = \frac{1}{2\pi C_2(R_1 + R_2)}$$

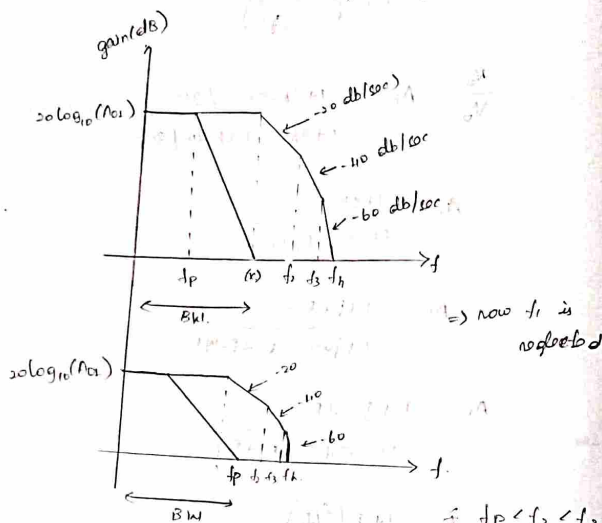
$$f_1 = \frac{1}{2\pi R_2C_2} \Rightarrow \text{frequency due to insertion of zero.}$$

To choose R_2 & C_2 value so that

the f_1 & f_p (1st corner frequency) should be same & it follows.

$$f_1 < f_p < f_2 < f_3 < f_h$$

$\therefore f_1 = f_1$ (due to zero insertion)
 $\Rightarrow$ so f_1 neglected.



$\Rightarrow$ Bandwidth is increased, compare 2 graphs.

Overall gain $= A \cdot A_1$

$$\frac{V_o'}{V_{in}} = \frac{A_o [1 + j(f/f_1)]}{[1 + j(f/f_1)] [1 + j(f/f_2)] [1 + j(f/f_3)] [1 + j(f/f_p)]}$$

$$\frac{V_o'}{V_{in}} = \frac{A_{o1}}{[1 + j(f/f_2)] [1 + j(f/f_3)] [1 + j(f/f_p)]}$$

By using this

$\Rightarrow$ improve the bandwidth.

$\Rightarrow$ cancel the already existing frequency

so it reduces multiple corner frequency.

⑥ Given that CMRR of an op-amp is 100 dB
 input common voltage is 15 V, differential
 voltage gain is 4000. calculate the
 output common voltage

Sol:-

$$\text{CMRR} = \frac{A_d}{A_c} = \frac{\text{differential mode gain}}{\text{common mode gain}}$$

$$(\text{CMRR})_{\text{dB}} = 20 \log_{10} \left[\frac{A_d}{A_c} \right] = 100 \text{ dB}$$

$$A_d = \frac{V_o}{V_d} = 4000$$

$$A_c = \frac{V_o}{V_c} ; V_c = 15 \text{ V}$$

$$20 \log_{10} \left[\frac{A_d}{A_c} \right] = 100$$

$$\log_{10} \left[\frac{A_d}{A_c} \right] = \frac{100}{20} = 5$$

$$\log_{10} \left[\frac{4000}{A_c} \right] = 5$$

$$\frac{4000}{A_c} = 10^5$$

$$A_c = \frac{4000}{10^5}$$

$$A_c = 0.04$$

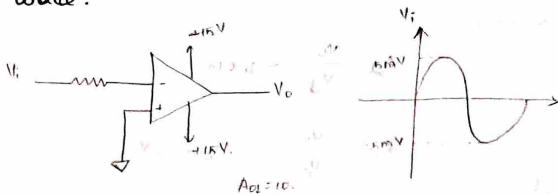
$$A_c = \frac{V_o}{V_e}$$

$$0.04 = \frac{V_o}{12}$$

$$V_o = 0.04 \times 12$$

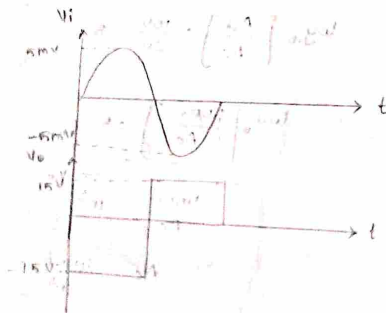
$$V_o = 0.48 \text{ Volt}$$

- ⑥. Examine the output voltage for the given circuit. Assume that the input V_i is 5mV sine wave.



Sol:

For ideal open loop op-amp, gain is ∞ .



It is called squarewave converter

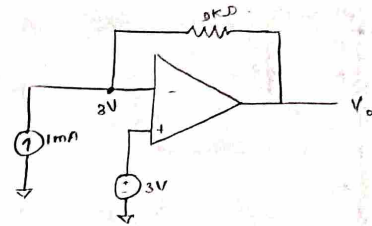
Assume $A_{OL} = 10^6$

$$V_o = A_{OL} \cdot V_{in}$$

$$= 5000 > 15V.$$

$$= -5000 < -15V.$$

- ⑦. Find the o/p for the given circuit.



2 steps to consider:

1. There is no current flow into the i/p terminal of op-amp
2. Use virtual ground concept.

Sol:

Nodal analysis:

$$\frac{-1 \times 10^{-3}}{5 \times 10^3} + \frac{3 - V_o}{5 \times 10^3} = 0$$

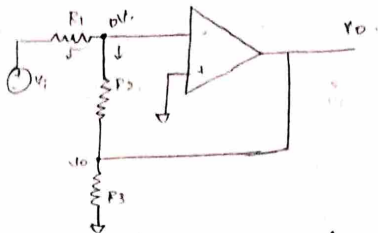
$$\frac{3 - V_o}{5 \times 10^3} = 1 \times 10^{-3}$$

$$3 - V_o = 5 \times 10^3 \times 10^{-3}$$

$$3 - 3 = V_o$$

$$V_o = 0V$$

- ⑧. Assume the op-amp to be ideal. Find the voltage gain of the amplifier shown below



- ⑨ An op-amp has an input offset voltage of 3mV , the closed loop gain is 100 . Find the output offset voltage.

Sol:

$$\frac{0 - V_i}{R_1} + \frac{0 - V_o}{R_2} = 0$$

$$\frac{-V_i}{R_1} = \frac{V_o}{R_2}$$

$$\therefore \frac{V_o}{V_i} = -\frac{R_2}{R_1} \quad \therefore \text{[inverting amplifier]}$$

⑩ Sol:

Given, $V_{OS} = 3\text{mV}$.

$$A = 100$$

$$V_o = A \cdot V_{OS} = 100 \times 3 \times 10^{-3}$$

$$V_o = 0.3\text{V}$$

$$V_o = 300\text{mV}$$

- ⑩ For an JCT41 op-amp, the i/p offset voltage changes by $30\mu\text{V}$ when the supply voltage

changes from $\pm 15\text{V}$ to $\pm 14\text{V}$. Find the value of PSRR and $(\text{PSRR})_{\text{dB}}$

Sol:

Given,

$$\text{PSRR} = \frac{\Delta V_{OS}}{\Delta V_S} ; \text{SVRR} = \frac{\Delta V_o}{\Delta V_S}$$

$$\Delta V_{OS} = 30\mu\text{V}$$

$$\Delta V_S = 15\text{V} - 14\text{V} = 1\text{V}$$

$$\text{PSRR} = \frac{30 \times 10^{-6}\text{V}}{1\text{V}} = 30\mu\text{V} \rightarrow \text{amount of error produced}$$

$$(\text{PSRR})_{\text{dB}} = 20 \log_{10} \left[\frac{1}{\text{PSRR}} \right]$$

$$= 90.457$$

$$\approx 90.46\text{dB}$$

- ⑩ An JCT41 op-amp configured in open-loop operation shows a change in o/p voltage of 6mV when the positive supply voltage changes by 2V while the i/p voltage is held constant. i) Calculate SVRR in mV/V

ii) Convert SVRR into dB.

Sol:

Given, $\Delta V_o = 6\text{mV}$.

$$\Delta V_S = 2\text{V}$$

$$1). \text{SVRR} = \frac{\Delta V_o}{\Delta V_S} = \frac{6\text{mV}}{2\text{V}} = \frac{3\text{mV}}{1\text{V}}$$

$$ii). (\text{SVRR})_{\text{dB}} = 20 \log_{10} \left[\frac{1}{3 \times 10^{-3}} \right]$$

$$\approx 50.46\text{dB}$$

Unit-3.

Applications of OP-AMP ~

2 types:

1. Linear application. [linear relationship b/w i/p & o/p]
2. Non-Linear application.

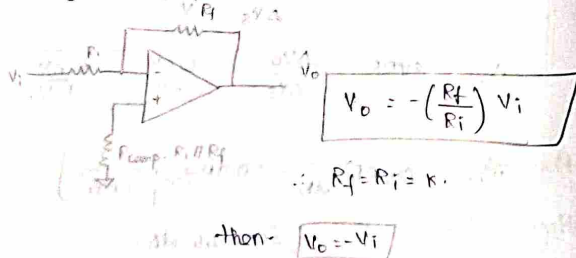
Linear

- current to voltage converter.
- voltage to current converter.
- adder.
- subtractor.
- summing amplifier.
- Instrumentation amplifier.

Non-Linear

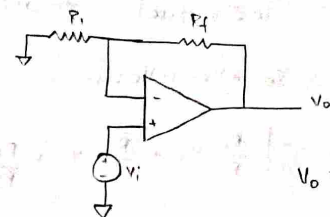
- Differentiator.
- Integrator.
- Rectifier.
- Low pass filter.
- high pass filter.
- peak detector.
- logarithmic amplifier.

①. Inverting amplifier.



$V_o = -V_i \rightarrow$ act as scale changer / inverter.

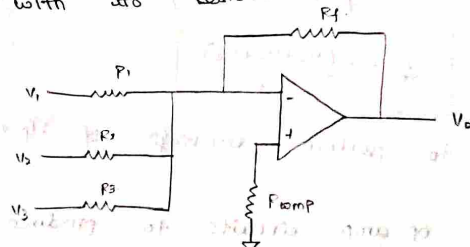
②. Non-inverting amplifier.



Inverting summing amplifier.

$\Rightarrow$ to get this, we modify the inverting amplifier.

* By applying multiple inputs along with its resistance.



* It performs addition and amplification.

$$V_{o1} \Rightarrow V_o \text{ due to } 'V_1' \text{ alone} = -\frac{R_f}{R_1} V_1$$

$$V_{o2} \Rightarrow \text{o/p } (V_o) \text{ due to } 'V_2' \text{ alone} = -\frac{R_f}{R_2} V_2$$

$$V_{o3} \Rightarrow V_o \text{ due to } 'V_3' \text{ alone} = -\frac{R_f}{R_3} V_3$$

by superposition theorem,
Total output is equal to
output of each individual source.

$$V_o = V_{o1} + V_{o2} + V_{o3}.$$

$$V_o = - \left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right]$$

if $R_1 = R_2 = R_3 = R_f$ it performs addition of i/p's.

$$\text{then, } V_o = - (V_1 + V_2 + V_3)$$

(it has 180° phase shift)
btw i/p & o/p.

if $R_1 = R_2 = R_3 = 3R_f$.

$$V_o = - \left[\frac{V_1}{3} + \frac{V_2}{3} + \frac{V_3}{3} \right]$$

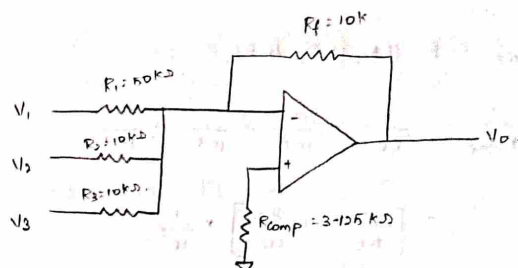
$$V_o = - \frac{(V_1 + V_2 + V_3)}{3}$$

It can able to perform average of i/p's.

① design an op-amp circuits to produce
the output expression of $V_o = - (0.2 V_1 + V_2 + V_3)$

Sol: Inverting summing amplifier.

$$V_o = - \left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right]$$



compare equation,

$$\frac{R_f}{R_1} V_1 = 0.2 V_1$$

$$\frac{R_f}{R_1} = 0.2$$

$$R_1 = \frac{R_f}{0.2}$$

Let choose $R_f = 10k$.

$$R_1 = \frac{10k}{0.2} = 50k\Omega$$

Similarly,

$$\frac{R_f}{R_2} V_2 = V_2$$

$$R_f = R_2$$

$$\therefore R_2 = 10k\Omega$$

$$\frac{R_f}{R_3} V_3 = V_3$$

$$R_3 = R_f$$

$$\therefore R_3 = 10k\Omega$$

$$R_{comp} = R_1 \parallel R_2 \parallel R_3 \parallel R_f$$

$$\frac{1}{R_{comp}} = \frac{1}{50k} + \frac{1}{10k} + \frac{1}{10k} + \frac{1}{10k}$$

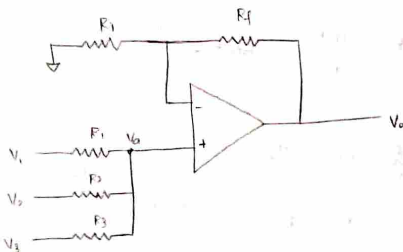
$$= \left[\frac{1}{50} + \frac{3}{10} \right] \times \frac{1}{10^3}$$

$$\frac{1}{R_{comp}} = \frac{16}{50} \times \frac{1}{10^3}$$

$$R_{comp} = \frac{50}{16} \times 10^3$$

$$R_{comp} = 3.125 \times 10^3 \Omega$$

Non Inverting - Summing Amplifier.



$$V_o = \left[1 + \frac{R_f}{R_1} \right] V_a \rightarrow (1)$$

Apply nodal analysis at V_a .

$$\frac{V_a - V_1}{R_1} + \frac{V_a - V_2}{R_2} + \frac{V_a - V_3}{R_3} = 0$$

$$V_a \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right] = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}$$

$$V_a = \frac{\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \rightarrow (2)$$

Substitute (2) in equ (1).

$$V_o = \left[1 + \frac{R_f}{R_1} \right] \frac{\left[\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right]}{\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right]}$$

Let us assume that

$$R_1 = R_2 = R_3 = R$$

$$V_o = \left[1 + \frac{R_f}{R_1} \right] \frac{\frac{V_1 + V_2 + V_3}{R}}{\frac{3}{R}}$$

$$V_o = \left(1 + \frac{R_f}{R_1} \right) \left(\frac{V_1 + V_2 + V_3}{3} \right)$$

choose, $R_f = \frac{R_1}{2}$

$$V_o = \left[1 + \frac{R_f}{R_1} \right] \left[\frac{V_1 + V_2 + V_3}{3} \right]$$

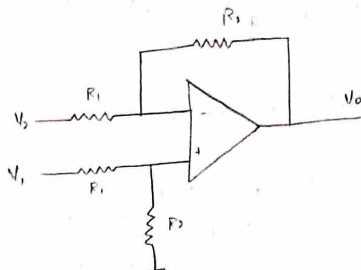
$$V_o = (2) \frac{V_1 + V_2 + V_3}{3}$$

$$\therefore V_o = V_1 + V_2 + V_3$$

$\therefore$ input and output of circuit

is same phase.

Subtractor:
Difference Amplifier is used here.



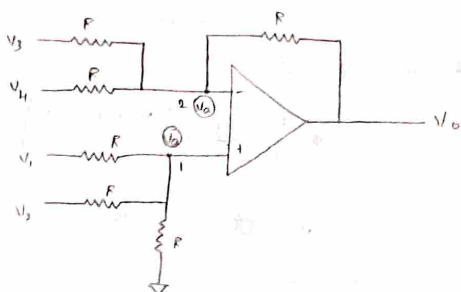
$$V_0 = \frac{R_3}{R_1} (V_1 - V_2)$$

if $R_3 = R_1$,

$$\text{then } V_0 = (V_1 - V_2)$$

$\Rightarrow$ It performs subtractor.

Adder - subtractor circuit:



Apply nodal analysis at node 1,

$$\frac{V_0 - V_1}{R} + \frac{V_0 - V_2}{R} + \frac{V_0 - 0}{R} = 0$$

$$\therefore V_0 \left[\frac{1}{R} + \frac{1}{R} + \frac{1}{R} \right] = \frac{V_1}{R} + \frac{V_2}{R}$$

$$V_0 = \frac{V_1 + V_2}{\frac{3}{R}}$$

$$V_0 \left[\frac{3}{R} \right] = \frac{V_1 + V_2}{R} \rightarrow \text{①}$$

Apply nodal analysis at node 2,

$$\frac{V_0 - V_3}{R} + \frac{V_0 - V_4}{R} + \frac{V_0 - V_0}{R} = 0$$

$$V_0 \left[\frac{1}{R} + \frac{1}{R} + \frac{1}{R} \right] = \frac{V_3}{R} + \frac{V_4}{R} + \frac{V_0}{R}$$

$$V_0 \left[\frac{3}{R} \right] = \frac{V_3 + V_4 + V_0}{R} \rightarrow \text{②}$$

equating ① & ②,

$$\frac{V_1 + V_2}{R} = \frac{V_3 + V_4 + V_0}{R}$$

$$\therefore V_0 = (V_1 + V_2) - (V_3 + V_4)$$

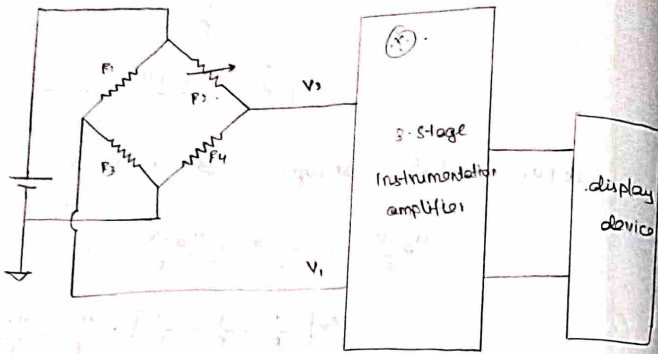
Instrumentation amplifier: [transducer bridge Based Instrumentation amplifier]

Transducer - convert non-electrical signal to electrical signal.

It is able to measure any physical quantity.

eg temp, humidity, water flow, pressure

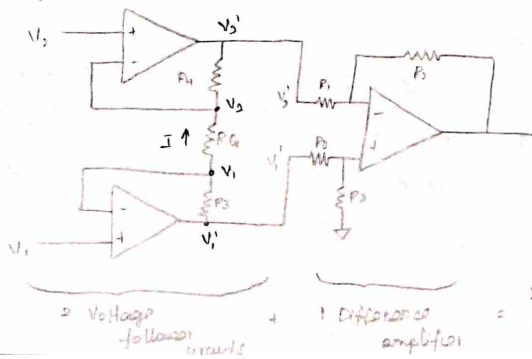
The o/p of transducer is very weak. It is connected to amplifier in that we can see the o/p.



V_2 & V_1 is weak so it connected to instrumentation amplifier.

* In difference amplifier loading effect is created becoz the input impedance are low to amplify.

* For instrumentation amplitude, the input sources given through the voltage follower circuit to the difference amplifier, by this loading effect is overcome.



Advantage:

1. It offers high gain accuracy.
2. It offers high CMRR.
3. offers high stability.
4. low offset voltages.
5. low output impedance.

$$V_o = \frac{R_2}{R_1} (V_1' - V_2') \rightarrow (1)$$

$$\text{here } I = \frac{V_1' - V_2'}{R_4} \Rightarrow V_1' - V_2' = I R_4 \rightarrow (2)$$

no current flows through input terminal

so, R_3, R_1, R_4 are connected in series

$$I = \frac{V_1' - V_2'}{R_3 + R_4 + R_4}$$

let, $R_3 = R_4$

$$I = \frac{V_1' - V_2'}{2R_3 + R_4} \rightarrow (3)$$

equate (1) & (3).

$$\frac{V_1 - V_2}{R_4} = \frac{V_1' - V_2'}{2R_3 + R_4}$$

$$\therefore V_1' - V_2' = \frac{2R_3 + R_4}{R_4} (V_1 - V_2)$$

$$V_1' - V_2' = \left[1 + \frac{2R_3}{R_4} \right] (V_1 - V_2) \rightarrow (4)$$

substitute (4) in eqn (1).

$$V_o = \left(\frac{R_3}{R_1} \right) \left(1 + \frac{R_3}{R_4} \right) (V_1 - V_2)$$

$\frac{R_3}{R_1} \rightarrow$ gain of difference amplifier.

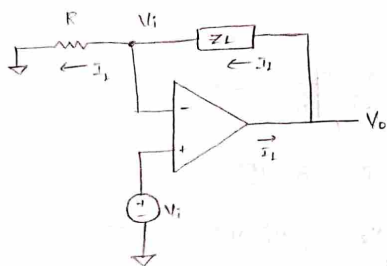
$1 + \frac{R_3}{R_4} \rightarrow$ gain of voltage follower amplifiers.

Voltage to Current Converter:
[Transconductance amplifier]

1. Floating load type V-I converter
2. Grounded load type V-I converter.

$$\Rightarrow \text{gain} = \frac{I}{V} = \frac{1}{R} = g_m$$

1. Floating load type V-I converter.



Load impedance is not grounded, it is floating, so connected to o/p.

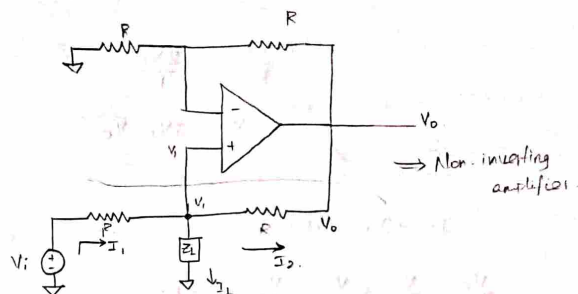
The applied voltage should decide the current.

$$I_1 = \frac{V_i}{R} \Rightarrow V_i = I_1 R$$

Precision resistor:

At any cost, value of impedance won't change.

2. Grounded load type V-I converter.



Here the load impedance is connected to ground.

Two steps are followed.

1. Apply KCL w.r.t V_i node.
2. Non-inverting amplifier.

$\therefore$ o/p of Non-inverting amplifier when V_i is known.

$$V_o = \left(1 + \frac{R}{R} \right) V_i$$

$$V_o = 2V_i \rightarrow \text{O}$$

Apply KCL eqn w.r.t Node ' V_i '

$$I_2 + I_1 = I_L$$

$$\frac{V_0 - V_i}{R} + \frac{V_i - V_i}{R} = \frac{V_i - 0}{Z_1}$$

Let us assume $Z_1 = R$.

$$\frac{V_0 - V_i}{R} + \frac{V_i - V_i}{R} = \frac{V_i}{R}$$

$$\frac{V_0 + V_i}{R} = \frac{2V_i}{R}$$

$$V_i = 2V_i - V_0$$

$$I_2 + I_1 = I_L$$

$$\frac{V_0}{R} - \frac{V_i}{R} + \frac{V_i}{R} - \frac{V_i}{R} = I_L$$

$$\frac{V_0 + V_i}{R} - \frac{2V_i}{R} = I_L$$

$$V_0 = I_L R + 2V_i - V_i \rightarrow \textcircled{1}$$

Sub $\textcircled{1}$ in $\textcircled{2}$.

$$V_0 = I_L R + \frac{2V_0}{2} - V_i$$

$$V_i = I_L R$$

$$I_L = \frac{V_i}{R}$$

Precision resistor

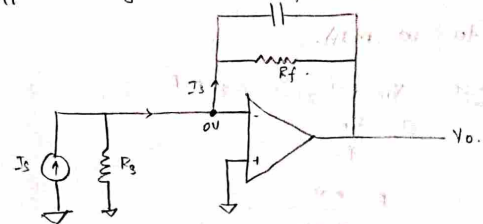
V-I:

* current produced, w.r to voltage.

circuit, which produces constant current through load impedance.

Current to Voltage Converter:

Applied input current will get o/p voltage.



assume, there is no feedback capacitance. $\therefore C_f \rightarrow$ eliminate high frequency noise signals.

apply KCL at o_v node.

$$I_s = \frac{0 - V_o}{R_f}$$

$$V_o = -I_s R_f$$

$$\text{gain} \Rightarrow \frac{V_o}{I_s} = -R_f = R_f$$

Gain of circuit expressed in resistance

$\therefore I_t$ is also known as transresistance amplifier.

C_f is not needed when we using low frequency signals.

- ③ Design a V-I converter with grounded load using an ideal op-amp such that i/p voltage range (0 to 5)V, output current range (0 to 10 mA).

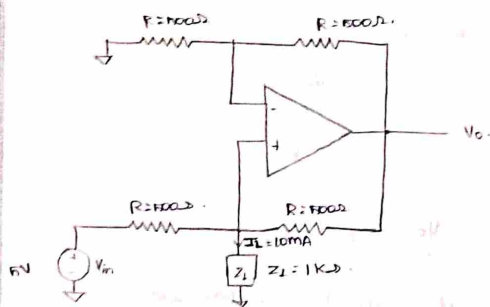
sol: Given, $V_{in} = 5V$, $I_L = 10mA$.

$$I_L = \frac{V_{in}}{R}$$

$$R = \frac{5V}{10 \times 10^{-3}}$$

$$R = 0.5 K\Omega$$

$$R = 500 \Omega$$



choose $Z_L = 1K\Omega$.

- ③ Design a grounded load V-I converter that produces o/p current is 2 mA/V of input signal, and maximum o/p voltage is 4 V.

sol:

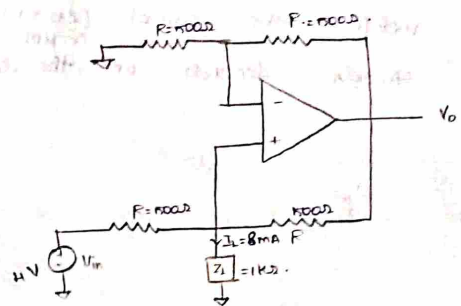
$$I_L = \frac{V_{in}}{R}$$

$$\frac{2mA}{1V} = \frac{I_L}{V_{in}} = \frac{1}{R}$$

$$R = \frac{1}{2 \times 10^{-3}}$$

$$R = 0.5 K\Omega$$

$$R = 500\Omega$$



$$\text{when } V_{in} = 1V \rightarrow I_L = 2mA$$

$$V_{in} = 4V \rightarrow I_L = 4 \times 2 \times 10^{-3}$$

$$I_L = 8mA$$

Resistance remains same. (that's why it is precision resistance).

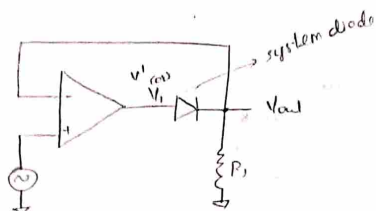
⇒ when we use the input less than cut-off voltage of diode ($< 0.7V$), the rectifier can't rectify. then we can go for op-amp based rectifier circuit which performs rectification.

Super diode:

whenever the combination of two diodes and op-amp.

Precision Rectifier:

Due to its open loop gain, the diode rectify the signal ($< 0.7V$) ($< \mu V$)
 V_{out} strongly depends on the diode



$$V_i = A(V_{in} - V_{out}) \quad (\text{or}) \quad V_i = A(V_i - V_o)$$

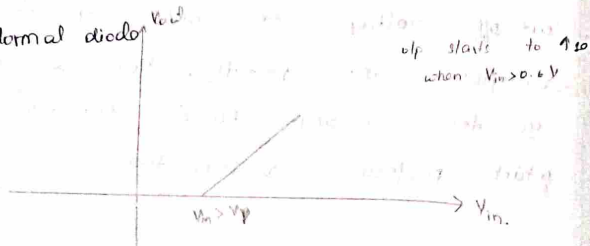
$$V_i = 0.7 + V_{out}$$

$$0.7 + V_{out} = A(V_{in} - V_{out})$$

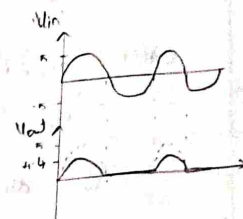
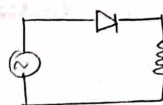
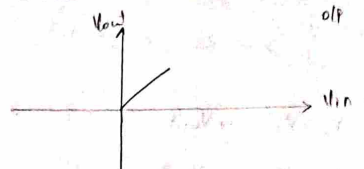
$$\frac{0.7 + V_{out}}{A} = V_{in} - V_{out}$$

for ideal diode $\rightarrow$ No knee voltage & built-in voltage.

1) Normal diode



2) Super diode:



for non-ideal diode
 * Usually, diode has internal built-in voltage of $0.6V$ - Silicon, $0.3V$ - Germanium. This diode is non-ideal diode.

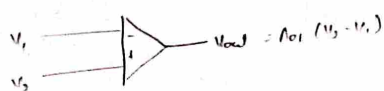
* It gives the o/p by reducing $0.6V$. It doesn't rectify when i/p is less than $0.6V$.

* In graph, it reduces the amplitude, whereas in ideal there is no change in amplitude.

* If we want o/p for $(i/p < 0.6V)$ then we use op-amp with diode.

$$\text{So, } V_{out} = \frac{0.6}{A_{ol}} = \frac{0.6}{10^4} = 60 \mu V$$

* Now at $60 \times 10^{-6} V$ the diode can rectify.



$$V_2 > V_1 \Rightarrow +V_{out}$$

$$V_2 < V_1 \Rightarrow -V_{out}$$

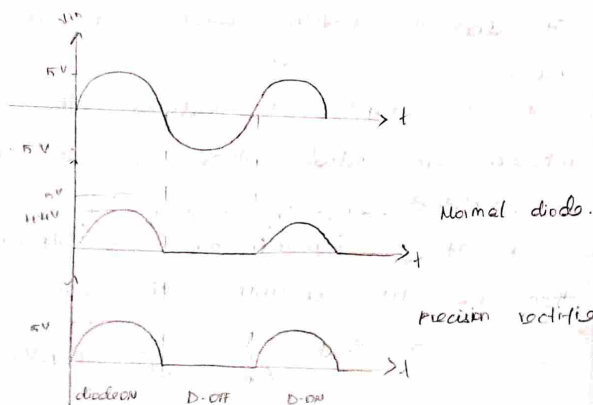
when, $V_1 < 0$.

$$V_1 = -V_{sat}$$

The diode is off & reverse bias

$$V_o = 0$$

In precision rectifier, it perform rectification with no drop voltage, in output.



Limitation:

*. It is only suitable to perform for low frequency signal. [due to it's slow rate it may not get o/p]

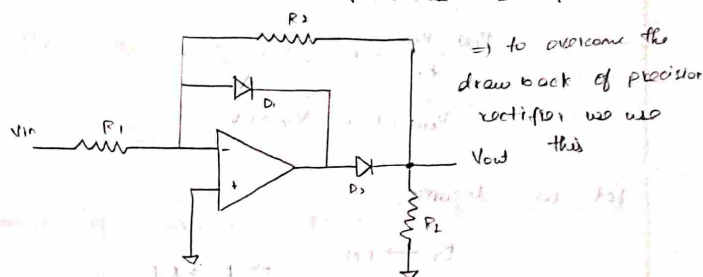
$V_{in} \rightarrow$ during +ve half cycle.

$$V_o' = +V_{sat}$$

$V_{in} \rightarrow$ during -ve half cycle.

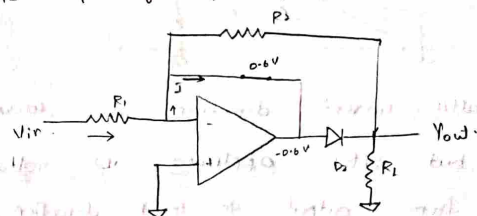
$$V_o' = -V_{sat}$$

modified precision half wave rectifier.



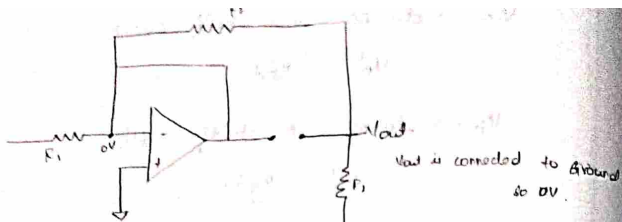
let as assume, $D_1 \rightarrow ON$, when $V_{in} > 0V$.

D_1 & D_2 is not directly connected to o/p of op-amp.



$D_2 \rightarrow OFF$ [beoz -0.6V to +ve terminal connected so reverse bias occurs].

Due to virtual ground concept -ve terminal has 0V. The circuit, $V_{out} = 0$.



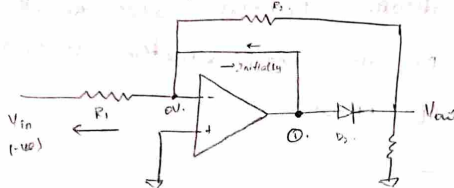
$$\frac{V_{out} - V_{in}}{R_2} = 0$$

$$V_{out} = 0, \quad V_{in} > 0V$$

Let us assume,

$D_1 \rightarrow ON, \Rightarrow D_1 \rightarrow R.B$

$V_{in} = -10V$



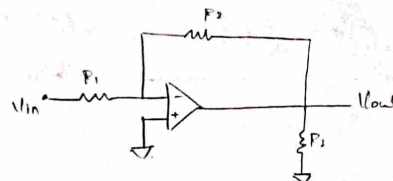
Initially current direction is towards V_{out} .

D-F.B but after applying -ve voltage I flows from output to input direction, direction changes, so D_1 is R.B, finally $D_1 \rightarrow OFF$.

$\therefore$ Voltage at 1 is greater than 0V (V_{in})

so +ve is connected to +ve of D_2 , D_2 - F.B, $D_2 \rightarrow ON$.

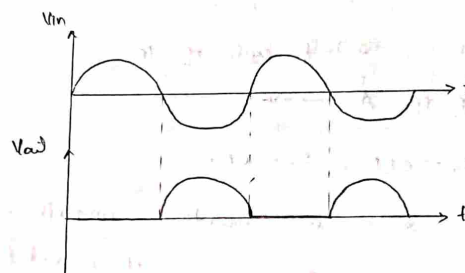
If $V_{in} = -10V$, it is inverting amplifier.



$$V_{out} = \left(-\frac{R_2}{R_1} \right) (-V_{in})$$

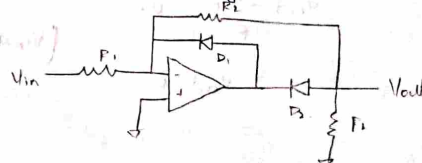
$$V_{out} = \frac{R_2}{R_1} V_{in}$$

Example of modified precision rectifier circuit.



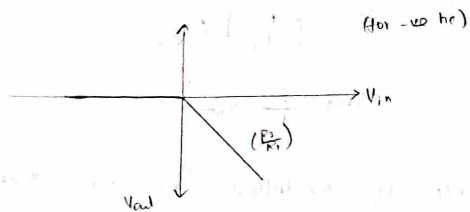
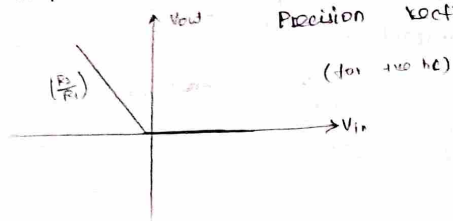
* only +ve h.c of circuit is rectified.

* If we want to rectify -ve half cycle, we change the bias of D_1, D_2



(It rectify $\odot$ -ve half cycle)

Input & output characteristics of Precision Rectifier.



$V_{in} > 0$, +ve half cycle of i/p.

o/p of 'A' $\rightarrow -ve$.

$D_1 \rightarrow \text{OFF}$, $D_2 \rightarrow \text{ON}$.

It is inverting amplifier.

$$V_o = -\frac{R_2}{R_1} V_{in} \quad \text{It is rectified with amplitude } \frac{R_2}{R_1}$$

$V_{in} < 0$, -ve half cycle of i/p.

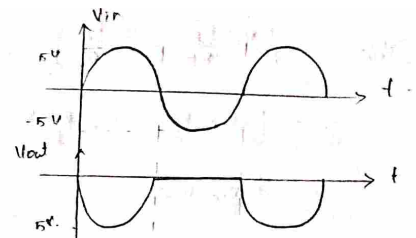
o/p of 'A' $\rightarrow +ve$.

$D_1 \rightarrow \text{ON}$, $D_2 \rightarrow \text{OFF}$.

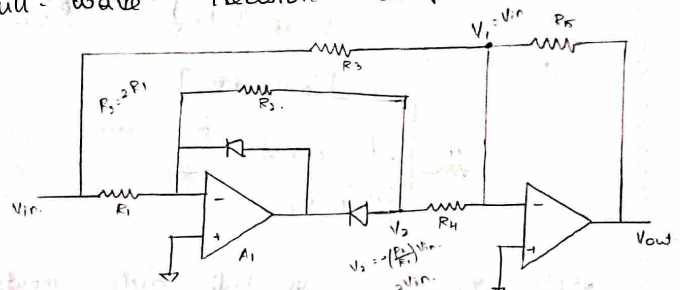
$$\frac{V_{out} - V_{in}}{R_2} = 0$$

$$V_{out} = 0$$

$$[V_{in} = 0]$$



Full-wave Precision Rectifier.



It is the combination modified half-wave rectifier followed by inverting summing amplifier.

When $V_{in} > 0$, [+ve half cycle i/p]

o/p of 'A1' $\rightarrow -ve$.

$D_1 \rightarrow \text{OFF}$ (R.B)

$D_2 \rightarrow \text{ON}$ (F.B).

$$V_o = V_s = -\frac{R_2}{R_1} V_{in} \quad (-ve)$$

$$V_i = V_{in} = +ve$$

When, $V_i = V_s$ with opp polarity.

$$R_1 = R_2 = R_4 = R$$

$$V_{out} = - \left(V_1 \frac{R_5}{R_4} + V_2 \frac{R_5}{R_3} \right)$$

$$= - \left(-V_{in} \frac{R_5}{R_4} + V_{in} \frac{R_5}{R_3} \right)$$

$$= - (-V_{in} + V_{in})$$

$$= 0$$

($R_5 = R_4 = R_3 = R$)

$$V_{out} = - \left[-V_{in} \left(\frac{R_5}{R_4} \right) + V_{in} \left(\frac{R_5}{R_3} \right) \right]$$

$$= - [-V_{in} + V_{in}]$$

$$= - (-V_{in})$$

$$\boxed{V_{out} = V_{in}}$$

when $V_{in} < 0$, -ve half cycle input.

o/p of 'A₁' $\rightarrow +V_{in}$.

$D_1 \rightarrow ON$, $D_2 \rightarrow OFF$

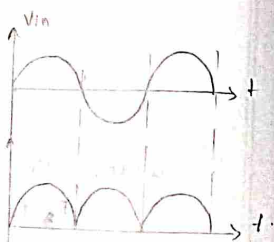
$$V_{out} = 0$$

$$\therefore V_2 = 0, \quad V_1 = -V_{in}$$

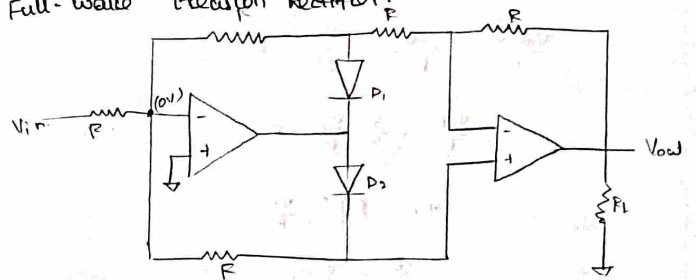
$$V_o = - \left[V_2 \left(\frac{R_5}{R_4} \right) + V_1 \left(\frac{R_5}{R_3} \right) \right]$$

$$= - [0 - V_{in}]$$

$$V_o = +V_{in}$$



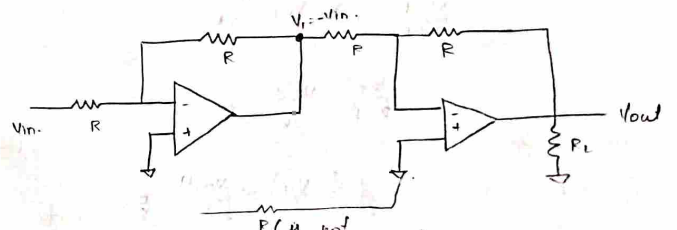
Full-wave Precision Rectifier:



$V_{in} > 0$, +ve half cycle of i/p.

$D_1 \rightarrow ON$ (short circuit)

$D_2 \rightarrow OFF$ (open circuit).



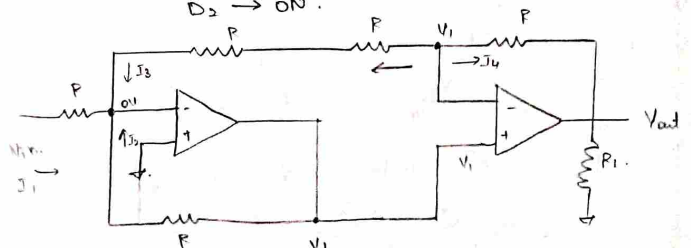
$$V_1 = V_{o1} = -V_{in}$$

$$V_{out} = - (-V_{in}) = V_{in}$$

$V_{in} < 0$, -ve half cycle of i/p.

$D_1 \rightarrow OFF$

$D_2 \rightarrow ON$.



$$\frac{V_{in}-0}{R} + \frac{V_1-0}{R} + \frac{V_1}{2R} = 0.$$

$$\frac{V_{in}}{R} + \frac{3}{2} \frac{V_1}{R} = 0.$$

$$V_{in} = -R \left(\frac{3}{2} \right) \frac{V_1}{R}.$$

$$V_{in} = -\frac{3}{2} V_1$$

$$V_1 = -\frac{2}{3} V_{in}.$$

$$I_3 + I_4 = 0.$$

$$\frac{V_1-0}{2R} + \frac{V_1-V_{out}}{R} = 0.$$

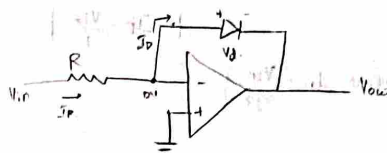
$$\frac{3V_1}{2R} - \frac{V_{out}}{R} = 0.$$

$$\frac{3}{2} \left(-\frac{2}{3} \frac{V_{in}}{R} \right) - \frac{V_{out}}{R} = 0.$$

$$\frac{-V_{in}}{R} = \frac{V_{out}}{R}.$$

$$V_{out} = -V_{in}.$$

Logarithmic Amplifier



$I_S \rightarrow$ reverse saturation current

$I_D \rightarrow$ diode current

$V_D \rightarrow$ diode voltage

$V_T \rightarrow$ Volt equivalent temperature

$$\text{diode current equation } I_D = I_S \left[e^{V_D/\eta V_T} - 1 \right]$$

①

where,

$$V_T = \frac{kT}{q}$$

$k \rightarrow$ boltzman's constant $= 1.38 \times 10^{-23} \text{ J/K}$

$q \rightarrow 1.602 \times 10^{-19} \text{ C}$ [coulomb]

$T \rightarrow$ temperature.

$$V_T = \frac{kT}{q} \approx 26 \text{ mV at room temp (25°C)}$$

from the given circuit.

$$0 = V_D + V_{out} = 0.$$

$$V_{out} = -V_D. \rightarrow \text{②}$$

$$\text{From equ ①, } \frac{I_D}{I_S} = \left[e^{V_D/\eta V_T} - 1 \right]$$

$$\frac{I_D}{I_S} + 1 = e^{V_D/\eta V_T}.$$

$$\therefore \frac{I_D}{I_S} \gg 1, \text{ neglect } 1.$$

$$\frac{I_D}{I_S} = e^{V_D/\eta V_T}$$

apply $\ln$ on b/s.

$$\ln \left(\frac{I_D}{I_S} \right) = \ln \left(e^{V_D/\eta V_T} \right)$$

$$V_D = \eta V_T \ln \left(\frac{I_P}{I_S} \right) \quad \left[\because I_P = \frac{V_{in}}{R} \right]$$

$$V_D = \eta V_T \ln \left(\frac{V_{in}}{I_S R} \right)$$

from equ ①,

$$V_{out} = -V_D$$

$$V_{out} = -\eta V_T \ln \left(\frac{V_{in}}{I_S R} \right)$$

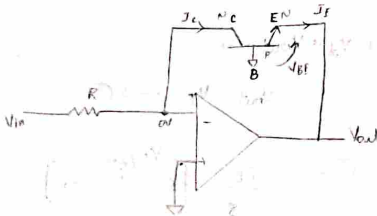
output $\propto$ logarithm of V_{in}

V_{out} depends on η , we don't need

this so we constructed transistor base logarithm

op-amp drawback: presence of η , dependence of temperature parameter

Transistor Base Log - Amplifier:



$$I_E = I_C + I_B \quad [\because \text{Base is grounded } I_B = 0]$$

$$I_E \approx I_C$$

collector current,

$$I_C = I_S \left[e^{V_{BE}/V_T} - 1 \right] \rightarrow ①$$

from the circuit.

$$0 + V_{BE} + V_{out} = 0$$

$$V_{out} = -V_{BE} \rightarrow ②$$

from equ ①,

$$\frac{I_C}{I_S} = \left[e^{V_{BE}/V_T} - 1 \right]$$

$$\frac{I_C}{I_S} + 1 = e^{V_{BE}/V_T}$$

$$\because \frac{I_C}{I_S} \gg 1, 1 \text{ is neglected.}$$

$$\frac{I_C}{I_S} = e^{V_{BE}/V_T}$$

apply natural log on b.s.

$$\frac{V_{BE}}{V_T} = \ln \left(\frac{I_C}{I_S} \right)$$

$$V_{BE} = \frac{I_P}{\beta_F} \rightarrow 0$$

$$\frac{V_{BE}}{V_T} = \ln \left(\frac{I_C}{I_S} \right) \quad \left[\because I_P = \frac{V_{in}}{R} \right]$$

$$V_{BE} = V_T \ln \left(\frac{V_{in}}{I_S R} \right) \rightarrow ③$$

sub ③ in ①.

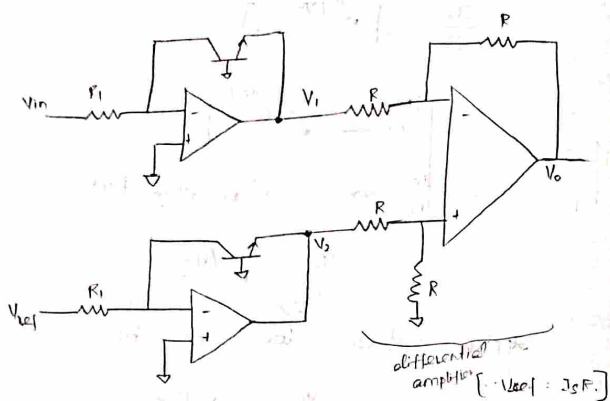
$$V_{out} = -V_T \ln \left(\frac{V_{in}}{I_S R} \right)$$

$$V_{out} = -\frac{kT}{q} \ln \left(\frac{V_{in}}{I_S R} \right)$$

Drawback:

Here, still V_{out} is dependent of temperature.

Temperature compensated transistor Based logarithmic amplifier.



$$V_1 = -\frac{KT}{q} \ln \left[\frac{V_{in}}{I_S R_1} \right]$$

$$V_2 = -\frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right]$$

By differential amplifier we know,

$$V_0 = +\frac{R_2}{R_1} (V_2 - V_1)$$

$$\therefore V_{out} = \frac{R}{R} (V_2 - V_1)$$

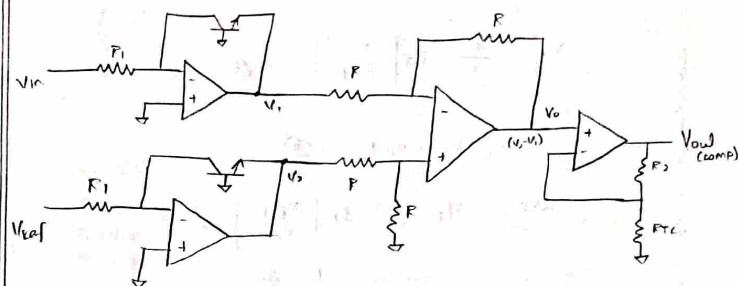
$$V_{out} = -\frac{KT}{q} \ln \left(\frac{V_{ref}}{I_S R_1} \right) + \frac{KT}{q} \ln \left(\frac{V_{in}}{I_S R_1} \right)$$

$$V_{out} = \frac{KT}{q} \ln \left[\frac{V_{in}}{V_{ref}} \right]$$

In order to compensate the T

we append one more circuit, as non-inverting

amplifier.

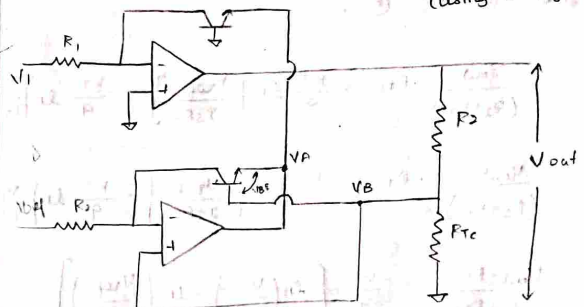


$R_2 \rightarrow$ temperature dependent resistor

$$V_{0(comp)} = \left(1 + \frac{R_2}{R_1} \right) \frac{KT}{q} \ln \left[\frac{V_{in}}{V_{ref}} \right]$$

when $T \uparrow \Rightarrow R_2$ also $\uparrow$ so both cancel each other, so temperature effect is cancelled.

Temperature compensated log-amplifier:
(using $\log$ -amp).



V_A - collector terminal of Transistor 1 &
collector terminal of Transistor 2.

$$V_A = -\frac{KT}{q} \ln \left[\frac{V_1}{I_S R_1} \right] \rightarrow (1)$$

$$\therefore V_B - V_E = V_A \rightarrow (3)$$

here, $V_E = \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] \Rightarrow$ derived from transistor base log-amp eqn
 $\rightarrow (2)$

sub (2) in (3).

$$V_B - \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] = -\frac{KT}{q} \ln \left[\frac{V_1}{I_S R_1} \right] \rightarrow (5)$$

here V_B = voltage across R_{TC}

by voltage divider theorem, $V_B = \frac{V_{out} \cdot R_{TC}}{(R_3 + R_{TC})}$

$\rightarrow (4)$

sub (4) in (5),

$$\frac{V_{out}}{(R_3 + R_{TC})} \cdot R_{TC} - \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] = -\frac{KT}{q} \ln \left[\frac{V_1}{I_S R_1} \right]$$

$$\frac{V_{out}}{(R_3 + R_{TC})} \cdot R_{TC} = \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] - \frac{KT}{q} \ln \left[\frac{V_1}{I_S R_1} \right]$$

$$\frac{V_{out} \cdot R_{TC}}{R_3 + R_{TC}} = -\frac{KT}{q} \left[\ln \left(\frac{V_1}{I_S R_1} \right) - \ln \left(\frac{V_{ref}}{I_S R_1} \right) \right]$$

V_B depends upon V_{out}

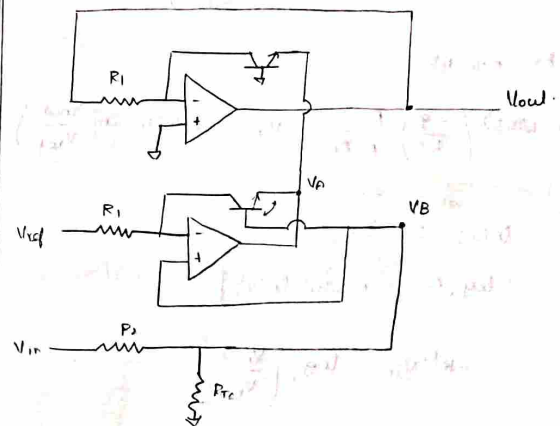
$$V_{out} = \left(\frac{R_3 + R_{TC}}{R_{TC}} \right) \left(-\frac{KT}{q} \right) \ln \left[\frac{\frac{V_1}{I_S R_1}}{\frac{V_{ref}}{I_S R_1}} \right]$$

$$\therefore V_{out} = \left[1 + \frac{R_3}{R_{TC}} \right] \left(-\frac{KT}{q} \right) \ln \left(\frac{V_1}{V_{ref}} \right)$$

This is how we can design a temperature compensated log-amplifier using only two logarithmic op-amp.

eg: $\log(1000) + 3$
 $\text{antilog}(10^3) = 1000$

Anti-log amplifier:



$$V_A = -\frac{KT}{q} \ln \left[\frac{V_{out}}{I_S R_1} \right] \rightarrow (1)$$

$$V_B - V_E = V_A$$

$$V_B = \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] = -\frac{KT}{q} \ln \left[\frac{V_{out}}{I_S R_1} \right]$$

$$\frac{V_{in} \cdot R_{TC}}{R_3 + R_{TC}} = \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] = -\frac{KT}{q} \ln \left[\frac{V_{out}}{I_S R_1} \right]$$

$$\frac{V_{in} \cdot R_{TC}}{R_3 + R_{TC}} = \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] - \frac{KT}{q} \ln \left[\frac{V_{out}}{I_S R_1} \right]$$

$$\frac{V_{in} R_{TC}}{R_3 + R_{TC}} = -\frac{KT}{q} \left[\ln \left(\frac{V_{out}}{I_S R_1} \right) - \ln \left(\frac{V_{ref}}{I_S R_1} \right) \right]$$

$$\left(\frac{-q}{KT} \right) \left(\frac{R_{TC}}{R_3 + R_{TC}} \right) V_{in} = \ln \left(\frac{V_{out}}{V_{ref}} \right)$$

xy by 0.4343,

$$(0.4343) \left(\frac{-q}{KT} \right) \frac{R_{TC}}{R_3 + R_{TC}} \cdot V_{in} = 0.4343 \ln \left(\frac{V_{out}}{V_{ref}} \right)$$

constants replace by k'

$$\therefore \ln(x) = 2.3 \log_{10}(x)$$

$$\log_{10}(x) = 0.4343 \ln(x)$$

$$-k' V_{in} = \log_{10} \left[\frac{V_{out}}{V_{ref}} \right]$$

$$\therefore \frac{V_{out}}{V_{ref}} = 10^{-k' V_{in}}$$

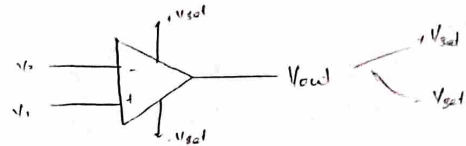
$$\therefore V_{out} = V_{ref} 10^{-k' V_{in}}$$

eg: If $k' = V_{in} = 1$ then.

$$V_{out} = \frac{V_{ref}}{10}$$

V_{ref} ↓ by 10 times.

COMPARATOR:



$$V_{out} = A_{OL} (v_1 - v_2) \quad [\because \text{If } (v_1 - v_2) \text{ is small}]$$

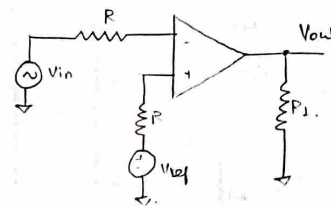
$$= +V_{sat} \text{ (or)} -V_{sat} \quad \text{due } A_{OL} (10^6) \text{ we get this}$$

It compare two terminal voltage, higher voltage terminal, the o/p go to corresponding V_{sat} .

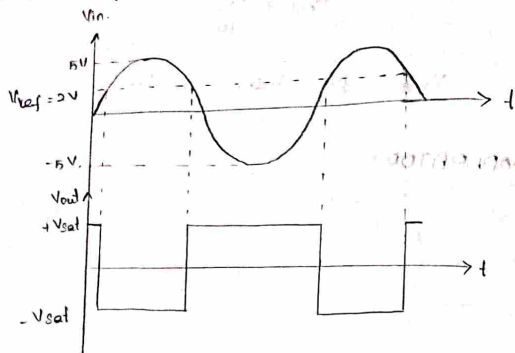
when, $V_1 > V_2$, o/p: $+V_{sat}$

$V_2 > V_1$, o/p: $-V_{sat}$.

Inverting mode comparator:



Assume, $V_{ref} = +V$.



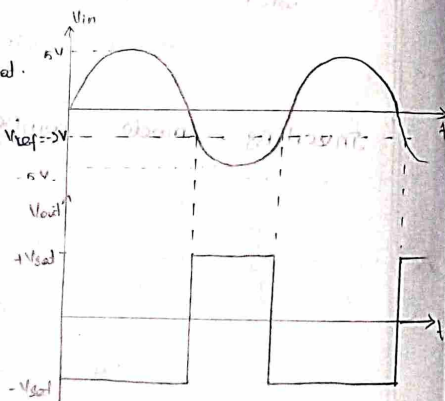
when, $V_{in} < V$, $V_{out} = +V_{sat}$.
($V_- < V_+$)

when, $V_{in} > V$, $V_{out} = -V_{sat}$.
($V_- > V_+$)

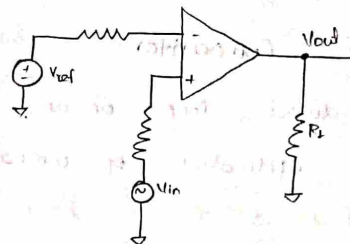
Assume, $V_{ref} = -V$.

when, $V_{in} > -V$, $V_o = -V_{sat}$.

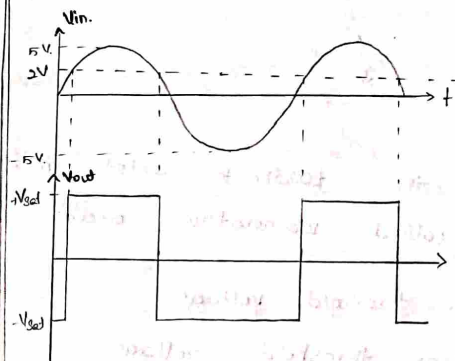
$V_{in} < -V$, $V_o = +V_{sat}$.



Non-Inverting Mode comparator.



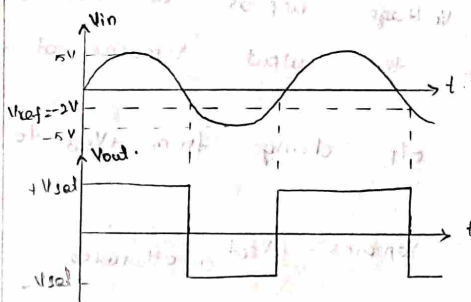
Assume, $V_{ref} = +V$.



$V_{in} < V \Rightarrow V_{out} = -V_{sat}$
 $V_- < V_+$

$V_{in} > V \Rightarrow V_{out} = +V_{sat}$
 $V_- > V_+$

Assume, $V_{ref} = -V$.



$V_{in} > -V \Rightarrow V_o = +V_{sat}$

$V_{in} < -V \Rightarrow V_o = -V_{sat}$
 $V_- < V_+$

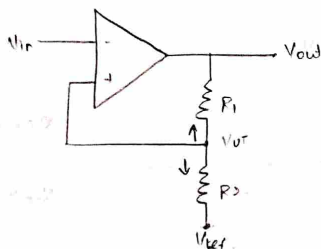
If we kept, $V_{sat} = 0V$ then it is zero crossing integrator circuit.

Bistable Regenerative Comparator:

⇒ Using closed loop op-amp.

⇒ It is application of comparator.

[also called as schmitt trigger]



whenever the feedback exist in the circuit is called regenerative circuit.

V_{UT} → upper threshold voltage.

V_{LT} → lower threshold voltage.

V_{UT} ⇒ Max Voltage appear at Non-inverting, [upto which the output remains at $+V_{sat}$]

(or) [when the o/p change from $+V_{sat}$ to $-V_{sat}$]

V_{LT} ⇒ output remains $-V_{sat}$, otherwise

It then

Minimum voltage, It remains $-V_{sat}$, otherwise the o/p transition from $-V_{sat}$ to $+V_{sat}$.

[Draw back of using open loop op-amp:

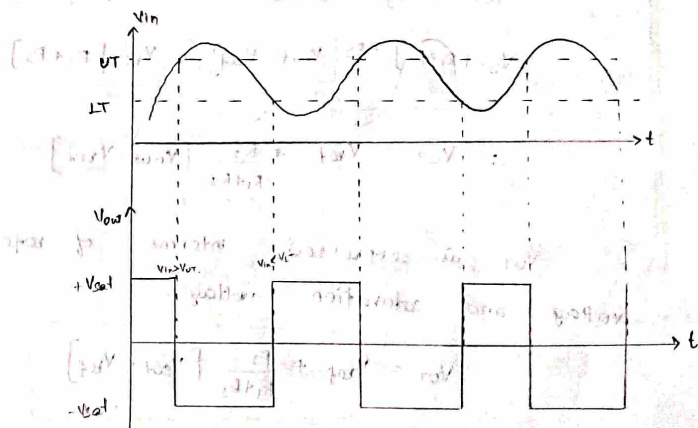
⇒ it has no threshold voltage]

⇒ o/p depends upon multiple threshold voltage [here we use 2].

when the i/p lies b/w V_{UT} & V_{LT} the o/p simply maintains its previous value. → hysteresis value.

V_{UT} → is a point max voltage at which the o/p change from $+V_{sat}$ to $-V_{sat}$.

V_{LT} → Max voltage at which o/p changes from $-V_{sat}$ to $+V_{sat}$.



when, $V_{out} = +V_{sat}$, $V_+ = V_{UT}$

apply nodal analysis, w.r.t. " V_{UT} "

$$\frac{V_{UT} - V_{out}}{R_1} + \frac{V_{UT} - V_{ref}}{R_2} = 0$$

$$V_{UT} \left[\frac{1}{R_1} + \frac{1}{R_2} \right] - \frac{V_{out}}{R_1} - \frac{V_{ref}}{R_2} = 0$$

$$V_{UT} \left[\frac{R_1 + R_2}{R_1 R_2} \right] = \frac{V_{out}}{R_1} + \frac{V_{ref}}{R_2}$$

$$V_{UT} [R_1 + R_2] = R_1 R_2 \left[\frac{V_{out}}{R_1} + \frac{V_{ref}}{R_2} \right]$$

$$V_{UT} [R_1 + R_2] = R_2 V_{out} + R_1 V_{ref}$$

add & sub by $R_2 V_{ref}$

$$V_{UT} [R_1 + R_2] = R_2 V_{out} + R_1 V_{ref} + R_2 V_{ref} - R_2 V_{ref}$$

$$V_{UT} [R_1 + R_2] = R_2 [V_{out} - V_{ref}] + V_{ref} [R_1 + R_2]$$

$$\therefore V_{UT} = V_{ref} + \frac{R_2}{R_1 + R_2} [V_{out} - V_{ref}]$$

V_{UT} is expressed in terms of reference voltage and saturation voltage.

$$V_{UT} = V_{ref} + \frac{R_2}{R_1 + R_2} [V_{sat} - V_{ref}]$$

when, $V_{out} = -V_{sat}$, $V_+ = V_{LT}$

$$\frac{V_{out} - V_{LT}}{R_1} + \frac{V_{ref} - V_{LT}}{R_2} = 0$$

apply nodal analysis w.r.t. V_{LT}

$$\frac{V_{LT} - V_{out}}{R_1} + \frac{V_{LT} - V_{ref}}{R_2} = 0$$

$$\frac{V_{LT} + V_{sat}}{R_1} + \frac{V_{LT} - V_{ref}}{R_2} = 0$$

$$V_{LT} \left[\frac{R_1 + R_2}{R_1 R_2} \right] = \frac{-V_{sat}}{R_1} + \frac{V_{ref}}{R_2}$$

xy by $R_1 R_2$:

$$V_{LT} [R_1 + R_2] = -R_2 V_{sat} + R_1 V_{ref} + R_2 V_{ref} - R_2 V_{ref}$$

$$V_{LT} [R_1 + R_2] = -R_2 [V_{sat} + V_{ref}] + V_{ref} [R_1 + R_2]$$

$$V_{LT} = \frac{-R_2}{R_1 + R_2} [V_{sat} + V_{ref}] + \frac{V_{ref} [R_1 + R_2]}{R_1 + R_2}$$

$$\therefore V_{LT} = V_{ref} - \frac{R_2}{R_1 + R_2} [V_{sat} + V_{ref}]$$

Hysteresis Voltage:

$$V_H = V_{UT} - V_{LT}$$

$$= \cancel{V_{ref}} + \frac{R_2}{R_1 + R_2} [V_{sat} - V_{ref}] - \cancel{V_{ref}} + \frac{R_2}{R_1 + R_2} (V_{sat} + V_{ref})$$

$$= \frac{R_2}{R_1 + R_2} [V_{sat} - V_{ref} + V_{sat} + V_{ref}]$$

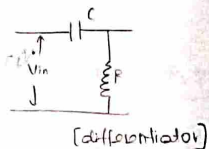
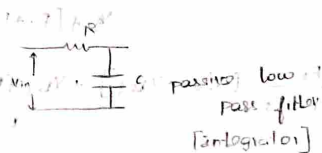
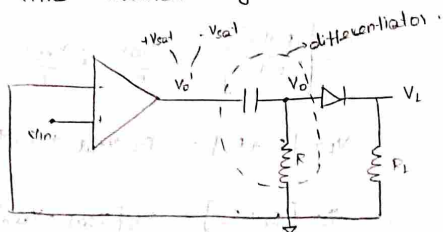
$$V_H = \frac{2 R_2}{R_1 + R_2} (V_{sat})$$

Advantage:

1. Noise immunity offered by regenerative comparator is high.
2. Switch speed is more.
3. Stability is more.

Application:

1. Time marker generative circuit.



$V_o = +V_{sat}$ $D \rightarrow ON$
 $V_o = -V_{sat}$ $D \rightarrow OFF$

MULT-VIBRATORS:

Types of multivibrators.

1. Astable $\rightarrow$ No stable states / both are quasi stable states.
2. Monostable $\rightarrow$ 1 stable state & 1 quasi stable state.
3. Bistable $\rightarrow$ 2 stable states.

Astable $\Rightarrow$ no need to apply external trip pulse to change its o/p states.

eg: oscillator, Relaxation oscillator.

- also called Free-Running multivibrator.

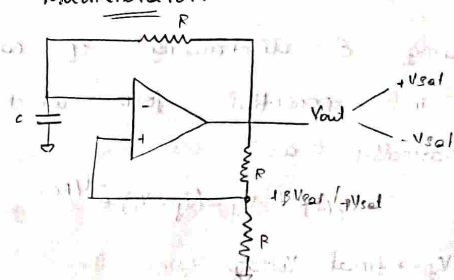
Monostable $\Rightarrow$ it requires -ve trigger pulse to change o/p from stable to quasi-stable state.

- also called one-shot multivibrator.

Bistable $\Rightarrow$ It requires two trigger pulse

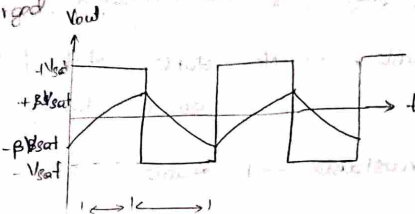
- also called binary circuit bcoz it has 2 stable state.

Astable Multivibrator:



→ Based on o/p

& discharged



→ o/p voltage depends upon threshold

voltage & charging/discharging of capacitor.

→ Initially Assume, $V_0 = +V_{sat}$, voltage at the terminal $+V_{sat}$.

→ capacitor → charged beyond β voltage

change $V_c > V_1$ then output is $-V_{sat}$

Assume, $V_0 = -V_{sat}$

$$V_1 = -\beta V_{sat}$$

→ discharge upto $-\beta V_{sat}$ when

it goes lower than $-\beta V_{sat}$

$V_c < V_1$ then the o/p is $+V_{sat}$

charging & discharging of capacitor

occur in exponential form, which is

mathematically as.

$$V_c(t) = V_f - (V_f - V_i) e^{-t/RC}$$

V_f → final voltage (Max voltage C can be charged is V_{sat}).

V_i → initial voltage.

T_1 — time period, when capacitor charged exponentially

T_2 — time period when capacitor discharged exponentially.

By Mathematical,

$$V_c(t) = V_{sat} - (V_{sat} - (-\beta V_{sat})) e^{-t/RC}$$

$$V_c(t) = V_{sat} - (V_{sat} + \beta V_{sat}) e^{-t/RC}$$

at $t = T_1$, $V_c(t) = +\beta V_{sat}$.

$$+\beta V_{sat} = V_{sat} - (V_{sat} + \beta V_{sat}) e^{-T_1/RC}$$

$$\beta V_{sat} - V_{sat} = -(V_{sat} + \beta V_{sat}) e^{-T_1/RC}$$

$$+V_{sat}(1 - \beta) = +V_{sat}(1 + \beta) e^{-T_1/RC}$$

$$\frac{1 - \beta}{1 + \beta} = e^{-T_1/RC}$$

Apply $\ln$ on B.S,

$$-T_1/RC = \ln\left(\frac{1 - \beta}{1 + \beta}\right)$$

$$\frac{T_1}{RC} = -\ln\left(\frac{1 - \beta}{1 + \beta}\right)$$

$$\therefore T_1 = RC \ln\left(\frac{1 + \beta}{1 - \beta}\right)$$

capacitor required same time for

charging & discharging.

If o/p is symmetrical,

$$T = T_1 + T_2 = 2T_1$$

$$T = 2RC \ln \left(\frac{1+\beta}{1-\beta} \right)$$

$\therefore$ Time period depends on R, C & β .

If we don't want V_{sat} in o/p we can use clipping circuit by zener diode.

connected (1 by 1 zener diode) in V_{out} , it limits the amplitude.

$$\beta = \frac{R_2}{R_1 + R_2}$$

$$\text{If } R_1 = R_2,$$

$$\beta = \frac{R_1}{R_1 + R_1} = \frac{1}{2} = 0.5$$

$$T = 2RC \ln \left[\frac{1+0.5}{1-0.5} \right] \Rightarrow T = 2RC \ln [3]$$

$$\text{If } R_1 = 1.16 R_2,$$

$$\beta = \frac{R_2}{1.16 R_2 + R_2} = \frac{1}{2.16} = 0.463$$

$$T = 2RC \ln \left[\frac{1+0.463}{1-0.463} \right] = 2RC \ln [2.724]$$

$$T = 2RC (1.000)$$

$$T_1 = 2RC$$

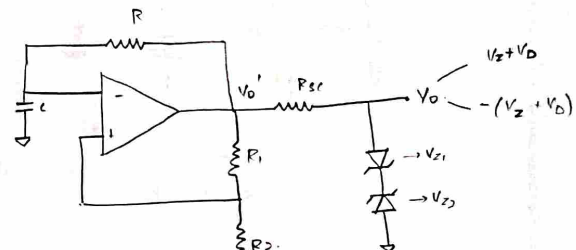
In this case time period is completely depends on external R, C .

$$\therefore f_0 = \frac{1}{T} = \frac{1}{2RC}$$

It is a symmetrical square wave generator circuit.

Symmetrical - on & off time is same.

For asymmetry square wave:



I am going to fix V_0 value by taking zener diode back to back. standard voltage of zener diode $\approx 5.6V$.

$R_{sc} \rightarrow$ current limiting resistor

(in order to prevent the breakdown of zener by allowing excessive current in $R \cdot B$).

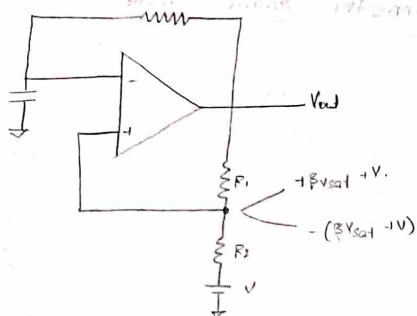
By connecting zener diode in parallel with load, the voltage across the load is $(V_Z + V_D) \approx (V_Z + V_D)$ rather than $+V_{sat} - V_{sat}$.

$$I = \frac{V_{out} - V_Z}{R_{ec}} \quad V_Z \rightarrow \text{breakdown voltage}$$

$V_{z1} = V_{z2} \Rightarrow$ generate symmetric square wave

$V_{z1} \neq V_{z2} \Rightarrow$ generate asymmetric " " " " " "

asymmetric square wave generator.



[application of stable :

*. Voltage to frequency conversion.]

$\Rightarrow$ We can control the frequency by changing the dc voltage.

⇒ The output of a table depends on external data supply.

Design a square wave oscillator for
 $f_o = 1 \text{ kHz}$ using an JCT41 op-amp and dc
 supply voltage of $\pm 15 \text{ V}$

Sol: Given, $f_0 = 1 \text{ kHz}$.

Design a symmetric square wave.

$$f_0 = \frac{1}{2RC}$$

$$R_1 = 1.16 R_2$$

Let $R_2 = 10\text{ k}\Omega$.

$$R_1 = 11.6 \text{ k}\Omega$$

$$f_0 = \frac{1}{2RC} = 1 \text{ kHz}$$

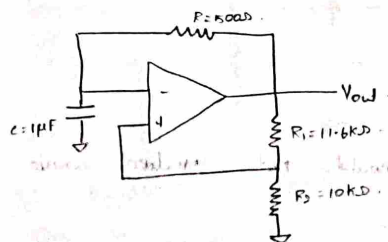
let us choose, $C = 1 \mu F$.

$$\frac{1}{2 \times 1 \times 10^{-8} \times 10^3} = R$$

$$\frac{1}{2 \times 10^{-3}} = R$$

$$R = 0.5 \text{ k}\Omega$$

$$R = 500 \Omega.$$



MONOSTABLE MULTIVIBRATOR:

1). It has 1 stable state & 1 quasi stable state.

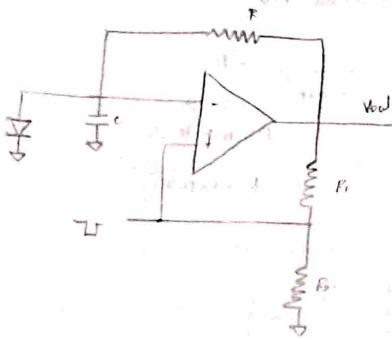
$\Rightarrow$ quasi stable state $\rightarrow$ Non-permanent state

2). It requires 1 triggering pulse externally. (to change state of op)

3). It is called 1-shot multi-vibrator.

4). Gating circuits - It can generate a pulse (or) square wave in specified time period (fixed time period).

5). Time delay circuit.



It should not produce wave form continuously.

By placing the diode in parallel to capacitor, when 0.7V the capacitor stop charging (so that it doesn't produce continuous)

$\rightarrow V_{out} = +V_{sat}$. (Before applying -ve trigger)

The voltage at capacitor fixed at ' V_0 '

Now, I need to apply with suitable amplitude. (giving -ve max voltage)

eg: $-4.00V < 0.00$
 $\downarrow$ $\downarrow$
 -ve trigger pulse. inverting terminal voltage.

[trigger $\rightarrow$ short duration pulse]

$\therefore$ Inverting terminal $>$ Non-inverting voltage, the [op $\rightarrow -V_{sat}$]

$\Rightarrow$ Now, capacitor start discharge upto $-V_{sat}$.

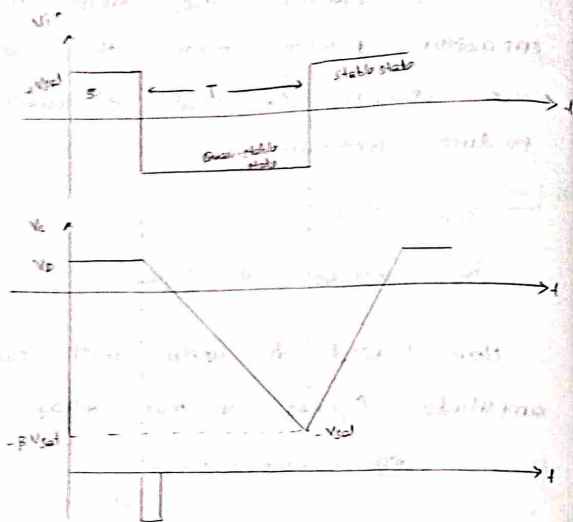
$\Rightarrow$ when it goes below $-V_{sat}$,

$V_- < V_+$ (-ve trigger) like $(-5.00 < -4.00)$,
 V_- V_+ (-ve trigger)

o/p $\rightarrow +V_{sat}$

$\Rightarrow$ due to in parallel connection of diode, capacitor discharge linearly.

after it charge upto $+V_{sat}$, but due to diode charge upto 0.7V after that remains V_0 .



$$V_c(t) = V_D - (V_D - V_i) e^{-t/RC}$$

$$V_c(t) = -V_{sat} - (-V_{sat} - V_D) e^{-t/RC}$$

$$\Rightarrow \therefore V_c(t) = -\beta V_{sat}, \text{ at } t = T$$

$$-\beta V_{sat} = -V_{sat} - (-V_{sat} - V_D) e^{-T/RC}$$

$$-\beta V_{sat} + V_{sat} = [V_{sat} + V_D] e^{-T/RC}$$

$$\frac{V_{sat} [1 - \beta]}{V_{sat} + V_D} = e^{-T/RC}$$

$$e^{T/RC} = \left[\frac{1 - \beta}{1 + \frac{V_D}{V_{sat}}} \right]$$

$$e^{T/RC} = \frac{1 + \frac{V_D}{V_{sat}}}{1 - \beta}$$

take $\ln$ on b/s,

$$\frac{T}{RC} = \ln \left(\frac{1 + \frac{V_D}{V_{sat}}}{1 - \beta} \right)$$

$$T = RC \ln \left(\frac{1 + \frac{V_D}{V_{sat}}}{1 - \beta} \right)$$

$\therefore V_D \rightarrow$ diode drop. (less $\approx 0.7V$).

$V_{sat} \rightarrow$ very high

$V_D < V_{sat}$, $\frac{V_D}{V_{sat}} \ll 1$, so it is neglected.

$$T = RC \ln \left[\frac{1}{1 - \beta} \right], \text{ here } \beta = \frac{R_2}{R_1 + R_2}$$

if $R_1 = R_2$,

$$\beta = \frac{R_2}{R_1 + R_2} = 0.5$$

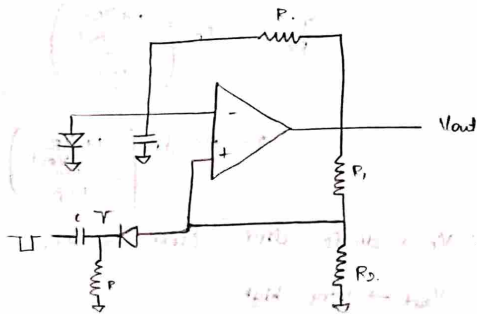
$$T = RC \ln \left[\frac{1}{1 - 0.5} \right] = RC \ln \left[\frac{1}{0.5} \right]$$

$$T = RC \ln(2)$$

$$\textcircled{\times} \quad T = 0.693 RC$$

By choosing RC , we can determine
generate a pulse of T .

diode is connected to ensure the
no trigger is applied, when the
trigger apply diode become OFF.



Q. Design a monostable multivibrator with
trigger pulse shaping which will drive a
LED ON for 0.5 sec, each time it is
pulsed.

Sol: Given, $T = 0.5 \text{ sec}$.

Formula, $T = 0.69 RC$.

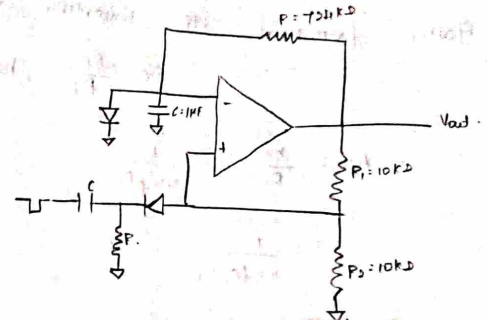
Let us choose $C = 1 \mu F$.

$$R = \frac{0.5}{0.69 \times 1 \times 10^{-6}}$$

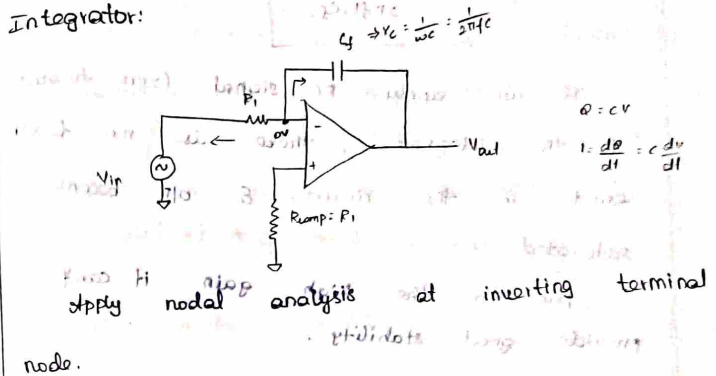
$$R = 0.7246 \times 10^6$$

$$R = 724.63 \text{ K}\Omega$$

Assume, $R_1 = R_2 = 10 \text{ K}\Omega$.



Integrator:



$$\frac{0 - V_{in}}{R_1} + \frac{0 - C \frac{dV_{out}}{dt}}{1} = 0$$

$$-\frac{C \frac{dV_{out}}{dt}}{1} = \frac{V_{in}}{R_1}$$

$$\frac{dV_{out}}{dt} = -\frac{V_{in}}{R_1 C_f}$$

$$\therefore V_{out} = -\frac{1}{R_1 C_f} \int V_{in} \cdot dt + V_c(0)$$

$V_c(0) \rightarrow$ initial voltage at capacitor

Gain, [compare with inverting amplifier]

$$A_c = -\frac{R_f}{R_i}, \text{ here } R_f = X_c$$

$$\therefore A_c = -\frac{X_c}{R_i} = -\frac{1}{\omega C_f R_i}$$

$$= -\frac{1}{2\pi f C_f R_i}$$

$$|A_c| = \frac{1}{2\pi f R_i C_f}$$

If we apply DC signal (zero frequency) to the integrator, there is no feedback exist in the circuit, & o/p becomes saturated. (becz it becomes open loop)

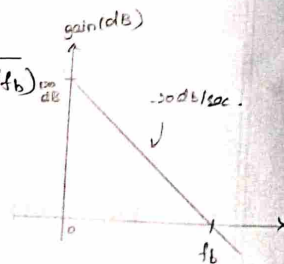
Due to the high gain it can't provide good stability.

open loop gain practical $\approx 10^6$

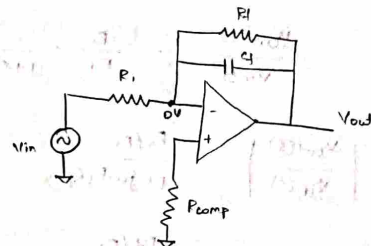
$$\begin{aligned} \text{gain in dB} &= 20 \log_{10}(10^6) \\ &= 20 \times 6 \\ &= 120 \text{ dB} \end{aligned}$$

$$|A_c| = \frac{1}{2\pi f R_i C_f}$$

$$f_b = \frac{1}{2\pi R_i C_f}$$



Practical Integrator: (Leaky integrator). To overcome the drawback we add one more resistor in feedback.



No, if we give 0 Hz as frequency, the impedance is ∞ , it becomes open circuit, [Now it is similar as inverting amplifier]. so o/p doesn't saturate. Now the gain is $-\frac{R_f}{R_i}$

Apply Nodal Analysis.

$$\frac{0 - V_{in}}{R_i} + \frac{0 - V_{out}}{R_f} - C \frac{dV_{out}}{dt} = 0$$

$$-C \frac{dV_{out}}{dt} = \frac{V_{in}}{R_i} + \frac{V_{out}}{R_f}$$

take Laplace transform.

$$-s C_f V_{out}(s) = \frac{V_{in}(s)}{R_i} + \frac{V_{out}(s)}{R_f}$$

$$\frac{V_{out}(s)}{R_f} + s C_f V_{out}(s) = -\frac{V_{in}(s)}{R_i}$$

$$V_{out}(s) \left[\frac{1}{R_f} + s C_f \right] = -\frac{1}{R_i} V_{in}(s)$$

cutoff frequency at which gain of circuit touch 0 dB.

$$V_{out}(s) \left[\frac{1+sR_fC_f}{R_f} \right] = -\frac{1}{R_i} V_{in}(s)$$

$$\frac{V_{out}(s)}{V_{in}(s)} = -\frac{R_f}{R_i} \cdot \frac{1}{1+sR_fC_f}$$

$$\left| \frac{V_{out}(s)}{V_{in}(s)} \right| = \frac{R_f/R_i}{1+j\omega R_fC_f}$$

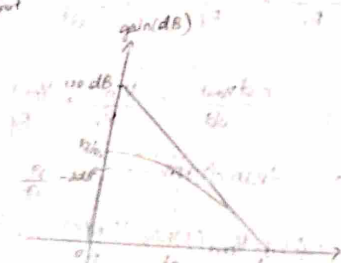
$$= \frac{R_f/R_i}{1+j\omega R_fC_f}$$

$$\left| \frac{V_{out}(s)}{V_{in}(s)} \right| = \frac{R_f/R_i}{1+j\omega R_fC_f}$$

$$\therefore f_0 = \frac{1}{2\pi R_f C_f}$$

At $f=0$, gain is $\frac{R_f}{R_i}$

When $f=f_0$, the gain is reduced to -3 dB.



when we design practical integrator we need to follow three rules:

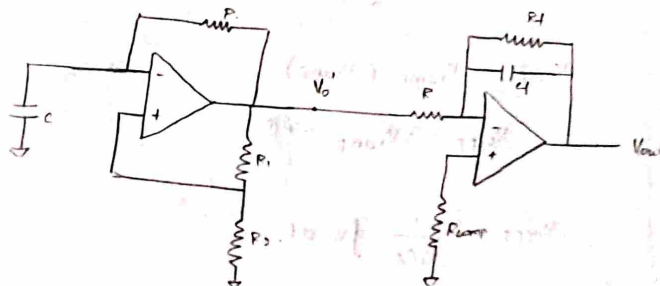
- ① Relationship between R_f & R_i , $\Rightarrow R_f = 10R_i$
- ② $f_{in} = 10 \cdot f_0$
- ③ $f_{oc} \leq f_L$

$\Rightarrow$ without R_f , C_f charge completely full

$\Rightarrow$ with R_f , amount of charging by capacitor is reduced, no possible for full (or) complete charge.

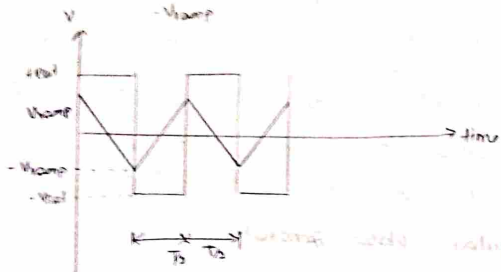
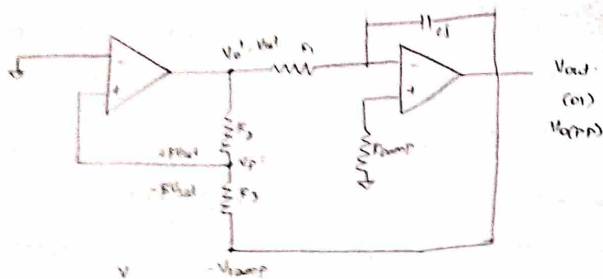
current divide & pass through R_f & C_f so leakage of current occurs. so it is called lossy. (loss of power due to R_f)

Triangular Wave generator:



In order to determine the frequency of square wave generator, remove R & C .

In order to change voltage of square of wave generator, connect output with R_f if.



$$V_o(pp) = V_{ramp} - (-V_{ramp})$$

$$V_o(pp) = 2V_{ramp} \rightarrow \text{---} \text{---} \text{---}$$

$$V_o(pp) = \frac{1}{R_1 C_f} \int V_0' dt$$

Initially we assume, $V_0' = +V_{sat}$

$$V_{out} = -V_{ramp}$$

Applying nodal analysis with respect to V_p

$$\frac{V_p - V_{sat}}{R_1} + \frac{V_p - (-V_{ramp})}{R_3} = 0$$

$$V_p \left[\frac{1}{R_1} + \frac{1}{R_3} \right] = \frac{V_{sat}}{R_1} - \frac{V_{ramp}}{R_3}$$

$$V_p \left(\frac{R_1 + R_3}{R_1 R_3} \right) = \frac{V_{sat}}{R_1} - \frac{V_{ramp}}{R_3}$$

$$V_p (R_1 + R_3) = R_3 V_{sat} - R_1 V_{ramp}$$

$$V_p (R_1 + R_3) = R_3 V_{sat} - R_1 V_{ramp} + R_3 V_{ramp} - R_3 V_{ramp}$$

$$V_p (R_1 + R_3) = R_3 (V_{sat} + V_{ramp}) - V_{ramp} (R_1 + R_3)$$

$$V_p = \frac{R_3}{R_1 + R_3} (V_{sat} + V_{ramp}) - V_{ramp}$$

$V_p \rightarrow$ is point at which the transition from $+V_{sat}$ to $-V_{sat}$

$$\text{make, } \therefore V_p = 0$$

$$\frac{R_3}{R_1 + R_3} [V_{sat}] + \frac{R_3}{R_1 + R_3} V_{ramp} - V_{ramp} = 0$$

$$\frac{R_3}{R_1 + R_3} \cdot V_{sat} + V_{ramp} \left[\frac{R_3}{R_1 + R_3} - 1 \right] = 0$$

$$V_{ramp} \left[\frac{R_3 - R_1 - R_3}{R_1 + R_3} \right] + \left(\frac{R_3}{R_1 + R_3} \right) V_{sat} = 0$$

$$R_3 V_{sat} = R_1 V_{ramp}$$

$$\therefore V_{ramp} = \frac{R_3}{R_1} V_{sat}$$

$$\therefore V_o(pp) = 2V_{ramp} = \frac{2R_3}{R_1} V_{sat} \quad \text{---} \text{---} \text{---}$$

$$V_{out}(pp) = \frac{1}{R_1 C_f} \int_0^{T/2} V_0' dt$$

$$= -\frac{1}{R_1 C_f} \int_0^{T/2} (-V_{sat}) dt$$

$$= \frac{1}{R_1 C_f} V_{sat} \cdot [t]_0^{T/2}$$

$$V_{out(pp)} = \frac{T \cdot V_{sat}}{2 R_1 C_f} \rightarrow (3)$$

equate (1) & (3)

$$\frac{2 R_3}{R_1} V_{sat} = \frac{T V_{sat}}{2 R_1 C_f}$$

$$T = \frac{4 R_1 R_3 C_f}{R_2}$$

This is time period of triangular wave generator.

$$\therefore f = \frac{1}{T} = \frac{R_2}{4 R_1 R_3 C_f}$$

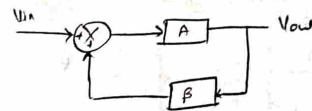
OSCILLATOR:

It is a electronic device, to produce a sinusoidal (or) non-sinusoidal signal periodically (continuously) without applying external input.

$$\text{-ve F.B} \rightarrow \frac{V_o(s)}{V_i(s)} = \frac{G}{1+GH} \quad \text{(or)} \quad \frac{A}{1+A\beta}$$

$$\text{+ve F.B} \rightarrow \frac{V_o(s)}{V_i(s)} = \frac{G}{1-GH} \quad \text{(or)} \quad \frac{A}{1-A\beta}$$

* crystal oscillator provides high stability.



Barkhausen criterion: $|AB| = 1$

$$\phi = 0^\circ \text{ or } 360^\circ$$

* If oscillator has -ve F.B, we won't get constant amplitude (it can't generate continuously)

* To design a oscillator, the circuit should have +ve feedback. to get continuously $\rightarrow$ should be in-phase.

If $A\beta < 1$, $A\beta > 1$; we won't get continuous wave without apply external i/p.

RC oscillators:

The feedback network is designed only with the help (use) of resistor & capacitor.

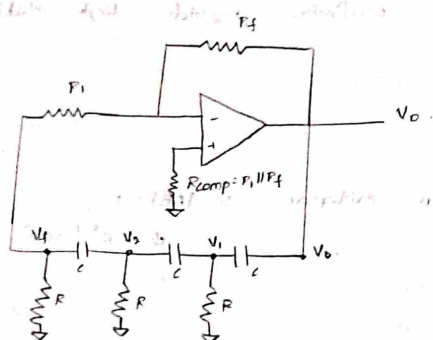
Drawback:

- * The o/p of signal the frequency is in Hz (or) kHz.
- * stability is low

①. R-C phase shift oscillator.

②. Wien-bridge oscillator.

③. RC phase shift oscillator.



op-amp provide -180° phase shift
feedback network - 180° phase shift
provide $= 360^\circ$

In F-B network, each RC network provide 60° phase shift to satisfy, $|A\beta| = 1$.

$$\beta = \frac{V_f}{V_o} \quad [\text{feedback gain}]$$

$$\therefore V_f = \beta V_o$$

Sustained oscillation:

It generates waveform with

constant amplitude & frequency.

$$|A\beta| = 1$$

If $|A\beta| < 1$, the amplitude decreases.
If $|A\beta| > 1$, the output becomes saturated.

write down KCL with respect to node V_1 .

$$\frac{V_1 - V_o}{X_c} + \frac{V_1 - 0}{R} + \frac{V_1 - V_2}{X_c} = 0$$

$$X_c = \frac{1}{sC}$$

$$(V_1 - V_o) sC + \frac{V_1}{R} + (V_1 - V_2) sC = 0$$

$$V_1 sC + \frac{V_1}{R} + V_1 sC - V_o(sC) - V_2(sC) = 0$$

$$V_1 \left[2sC + \frac{1}{R} \right] - V_o(sC) - V_2(sC) = 0$$

$$V_1 \left[\frac{1 + 2sCR}{R} \right] - V_o(sC) - V_2(sC) = 0$$

$$V_1 (1 + 2sCR) - V_o(RsC) - V_2(sCR) = 0$$

$$V_1 [1 + 2sCR] - V_2[sCR] = V_o[sCR]$$

$$V_o = \frac{V_1(1 + 2sCR)}{sCR} - V_2 \rightarrow \text{①}$$

write down KCL with respect to

node V_2

$$\frac{V_2 - V_1}{X_c} + \frac{V_2 - 0}{R} + \frac{V_2 - V_3}{X_c} = 0$$

$$(V_2 - V_1) sC + \frac{V_2}{R} + (V_2 - V_3) sC = 0$$

$$V_2 sC + \frac{V_2}{R} + V_2 sC - V_1(sC) - V_f(sC) = 0.$$

$$V_2 \left[2sC + \frac{1}{R} \right] - V_1(sC) - V_f(sC) = 0.$$

$$V_2 \left[\frac{1+2sCR}{R} \right] - V_1(sCR) - V_f(sCR) = 0.$$

$$V_f(sCR) = V_2(1+2sCR) - V_1(sCR)$$

$$V_f = \frac{V_2(1+2sCR)}{sCR} - V_1 \rightarrow \textcircled{D}$$

write down KCL with respect to node 2, V_f

$$\frac{V_f - V_2}{R_C} + \frac{V_f - 0}{R} = 0.$$

$$(V_f - V_2)sC + \frac{V_f}{R} = 0.$$

$$V_f \left[sC + \frac{1}{R} \right] - V_2(sC) = 0.$$

$$V_f \left[\frac{1+RSC}{R} \right] - V_2(sC) = 0.$$

$$V_f(1+RSC) - V_2(sCR) = 0.$$

$$V_2 = \frac{V_f(1+RSC)}{sCR} \rightarrow \textcircled{B}$$

Sub $\textcircled{B}$ in $\textcircled{D}$.

$$V_f = \frac{V_f(1+RSC)}{sCR} \cdot \frac{(1+2sCR)}{sCR} - V_1$$

$$V_f = \frac{V_f(1+RSC)(1+2sCR)}{s^2C^2R^2} - V_1$$

$$V_f = \frac{V_f(1+2sCR + sCR + 2s^2R^2C^2)}{s^2R^2C^2} - V_1$$

$$-V_f + \frac{V_f(1+3sCR + 2(sCR)^2)}{(sCR)^2} = V_1$$

$$\frac{-V_f(sCR)^2 + V_f(1+3sCR + 2(sCR)^2)}{(sCR)^2} = V_1$$

$$\frac{V_f[-(sCR)^2 + 1 + 3sCR + 2(sCR)^2]}{(sCR)^2} = V_1$$

$$V_1 = \frac{V_f[(sCR)^2 + 1 + 3sCR]}{(sCR)^2}$$

$$V_1 = \frac{V_f[1 + 3sCR + (sCR)^2]}{(sCR)^2}$$

sub V_1, V_2 in $\textcircled{D}$.

$$V_0 = \frac{V_f[1 + 3sCR + (sCR)^2]}{(sCR)^2} \cdot \frac{(1+2sCR)}{(sCR)} - \frac{V_f(1+RSC)}{sCR}$$

$$= \frac{V_f(1+3sCR + (sCR)^2)(1+2sCR)}{(sCR)^3} - \frac{V_f(1+RSC)}{sCR}$$

$$= \frac{V_f[1 + 3sCR + (sCR)^2](1+2sCR) - V_f(1+RSC)(sCR)^2}{(sCR)^3}$$

$$= \frac{V_f[1 + 3sCR + 3sCR + 6(sCR)^2 + (sCR)^2 + 2(sCR)^3] - V_f[(sCR)^2 + (sCR)^3]}{(sCR)^3}$$

$$V_o = \frac{V_i [1 + 5sRC + 7(sRC)^2 + 2(sRC)^3] - V_i [(sRC)^2 + (sRC)^3]}{(sRC)^3}$$

$$V_o = \frac{V_i [1 + 5sRC + 7(sRC)^2 + 2(sRC)^3 - (sRC)^2 - (sRC)^3]}{(sRC)^3}$$

$$V_o = \frac{V_i [1 + 5sRC + 6(sRC)^2 + (sRC)^3]}{(sRC)^3}$$

$$\frac{V_f}{V_o} = \frac{(sRC)^3}{[1 + 5sRC + 6(sRC)^2 + (sRC)^3]}$$

$$\beta = \frac{(sRC)^3}{(sRC)^3 \left[\frac{1}{(sRC)^3} + 5 \frac{1}{(sRC)^2} + \frac{6}{sRC} + 1 \right]}$$

$$\beta = \frac{V_f}{V_o} = \left[\frac{1}{1 + \frac{5}{(sRC)^2} + \frac{6}{sRC} + \frac{1}{(sRC)^3}} \right]$$

Replace s with $j\omega$ and ω with

$2\pi f$.

$$\beta = \frac{1}{1 + \frac{6}{j2\pi fRC} + \frac{5}{4\pi^2 f^2 R^2 C^2} + \frac{1}{(2\pi fRC)^3}}$$

$$\beta = \frac{1}{\left[1 - 5 \left(\frac{f}{f_0} \right)^2 \right] + \frac{1}{5} \left[6 \left(\frac{f}{f_0} \right) - \left(\frac{f}{f_0} \right)^3 \right]}$$

$$f_0 = \frac{1}{2\pi RC}$$

$$= \frac{1}{\left[1 - 5 \left(\frac{f}{f_0} \right)^2 \right] - \frac{1}{5} \left[6 \left(\frac{f}{f_0} \right) - \left(\frac{f}{f_0} \right)^3 \right]}$$

The imaginary term is zero.

$$\beta = \frac{1}{\left(1 - 5 \left(\frac{f}{f_0} \right)^2 \right)} \rightarrow (3)$$

$$\left[6 \left(\frac{f}{f_0} \right) - \left(\frac{f}{f_0} \right)^3 \right] = 0$$

$$6 \left(\frac{f}{f_0} \right) = \left(\frac{f}{f_0} \right)^3$$

$$\frac{f}{f_0} = \sqrt{6} \rightarrow (4)$$

$$f_0 = \frac{1}{\sqrt{6} 2\pi RC}$$

sub (11) in (3),

$$\beta = \frac{1}{1 - 5(6)}$$

$$= \frac{1}{1 - 30} = \frac{1}{-29}$$

$$A_v \cdot \beta = 1$$

$$A_v \cdot \left(-\frac{1}{29} \right) = 1$$

$$A_v \geq -29$$

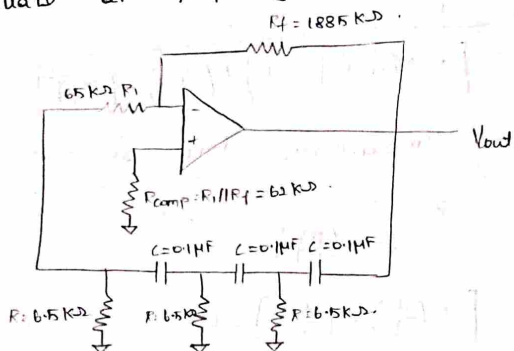
To find R_1 & R_f .

$$-\frac{R_f}{R_1} = -29$$

$$R_f = 29 R_1$$

$$R_1 = 10 R$$

6). Design an RC phase shift oscillator to oscillate at frequency $f_0 = 100\text{Hz}$.



$$f_0 = \frac{1}{\sqrt{6} \times \pi R C}$$

Let $C = 0.1 \mu\text{F}$.

$$100 = \frac{1}{\sqrt{6} \times 2 \times 3.14 \times R \times 0.1 \times 10^{-6}}$$

$$100 = \frac{1}{1.5385 \times 10^{-6} \times R}$$

$$\frac{1}{R} = 153.85 \times 10^{-6}$$

$$R = 6.501 \times 10^{-6} \times 10^{-3}$$

$$R = 6.5 \text{ k}\Omega$$

$$R_1 = 10(R)$$

$$= 10 \times 6.5 \times 10^{-3}$$

$$R_1 = 65 \text{ k}\Omega$$

$$R_f = 29(65 \text{ k}\Omega)$$

$$R_f = 1885 \text{ k}\Omega$$

$$\frac{1}{R_{\text{comp}}} = \frac{1}{R_1} + \frac{1}{R_f}$$

$$\frac{1}{R_{\text{comp}}} = \frac{1}{65000} + \frac{1}{1885000}$$

$$\frac{1}{R_{\text{comp}}} = \frac{1885000 + 65000}{(65000)(1885000)}$$

$$= \frac{1950000}{1.235 \times 10^{11}}$$

$$= \frac{1950 \times 10^3}{1.235 \times 10^{11}}$$

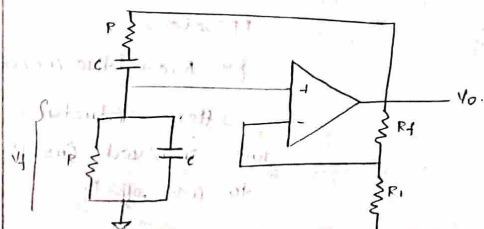
$$\frac{1}{R_{\text{comp}}} = 1591.83 \times 10^{-8}$$

$$R_{\text{comp}} = 6.282 \times 10^{-4} \times 10^8$$

$$= 6.282 \times 10^4$$

$$R_{\text{comp}} = 62 \text{ k}\Omega$$

WHEATSTONE BRIDGE OSCILLATOR:



Active Filters:

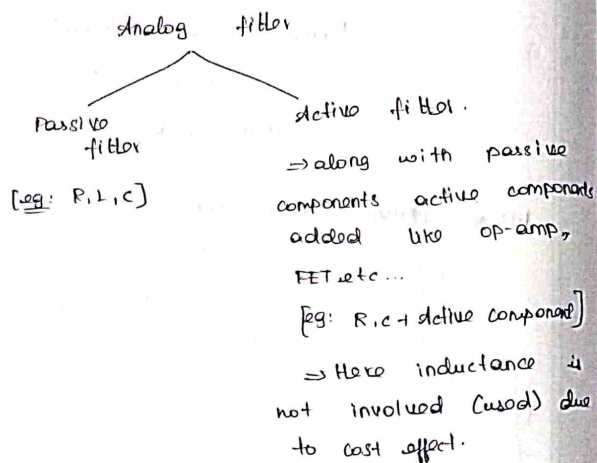
Filter $\rightarrow$ Frequency selective Network.

* Based on nature of signals as i/p.

Filter is classified in.

1. Analog filter.
2. Digital filter.

* In analog filter is classified based on the components used.



* Active filter based on suppression of i/p signal is classified as,

1. Low pass filter - LPF
2. High pass filter - HPF
3. Band pass filter - BPF.

1). BPF / BRF / BRF

* Based on frequency response, the filter classification.

reduction in gain occurs in monotonically also. $P \rightarrow$ Pass band, $SB \rightarrow$ Stop band.

1. Butterworth filter - flat PB, flat SB.

[You may not find the ripple]

2. Chebyshev filter $\uparrow$

3. Bessel function.

$\Rightarrow$ We use Butterworth filters.

Parameters:

- 1). Pass Band.

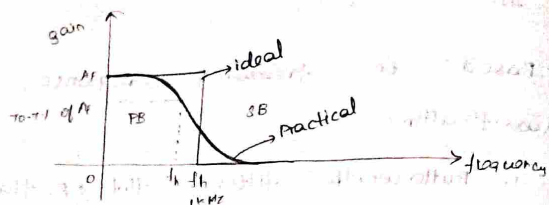
- 2). Stop Band.

- 3). Cut-off frequency.

Let us consider low pass filter.

f_c $\rightarrow$ maximum range of freq. that is transmitted by LP filter.

stop band $\rightarrow$ Range of frequency of i/p signal that can be suppressed/eliminated/attenuated by filter circuit.



For ideal characteristics of LPF,

PB = 100% gain. [we get exact transition]
SB = No gain.

For Practical,

* We can't get exact transition from high to low.

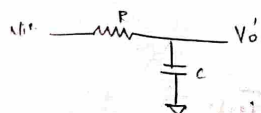
* It is monotonically decreasing (gain) as frequency is.

f_h → frequency at which the separation (of) transition from PB to SB: [ideal].
[cutoff / corner freq]

f_h → It is frequency at which the gain reduced by 2 dB.

The gain which is reduced to 70.7% of its maximum gain, that frequency is cut-off frequency. [Practical].

First order butlerworth active LPF:



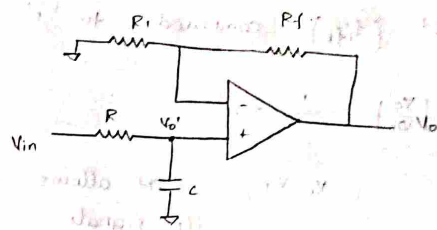
$$V_o' = \frac{V_{in}}{R + 1/sC} = \frac{1}{sC} V_{in}$$

Passive LPF

→ max gain is, $A = 1$.
when $(\omega = 0)$

$$A = \frac{V_o'}{V_{in}} = \frac{1}{1 + j\omega RC}$$

→ when $\omega = \infty$, $A = 0$.



$$V_o = \left(1 + \frac{R_f}{R_i}\right) V_o' \rightarrow \text{①}$$

Advantage:

→ In order to improve the gain of passive LPF we added active component. that advantage of active component is applied to active filter.
→ By choosing proper R_f & R_i

we can improve the gain.

$$V_o = A_f \cdot \frac{V_{in}}{1 + j\omega RC}$$

Advantage: 2nd

1. No loading effect in active filter
2. small size compared to passive filter.
3. low cost because inductor is not involved.

$$\frac{V_o}{V_{in}} = \frac{A_f}{1 + j(f/f_h)} \rightarrow \text{②}$$

where, $f_h = \frac{1}{2\pi RC}$.

$$\left| \frac{V_o}{V_{in}} \right| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_h} \right)^2}}$$

case (i): $f \ll f_h$.

Neglect $\left(\frac{f}{f_h} \right)$ compared to '1'

$$\left| \frac{V_o}{V_{in}} \right| = \frac{1}{1} = 1.$$

$V_o = V_{in}$. $\therefore$ It allows all the i/p signals.

case (ii): $f \gg f_h$.

neglect 1, $1 + \left(\frac{f}{f_h} \right)^2 \approx \left(\frac{f}{f_h} \right)^2$

$$\left| \frac{V_o}{V_{in}} \right| = \frac{1}{\sqrt{\left(\frac{f}{f_h} \right)^2}} = \frac{1}{f/f_h}$$

$$\left| \frac{V_o}{V_{in}} \right| = \frac{f_h}{f}$$

$\therefore$ signal is get attenuated (decreased).

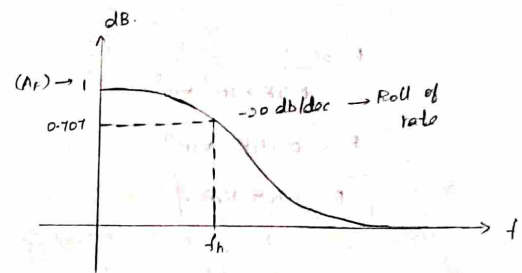
$\therefore$ It suppress the all i/p signal.

case (iii): $f = f_h$.

$$\left| \frac{V_o}{V_{in}} \right| = \frac{1}{\sqrt{1 + (1)^2}} = \frac{1}{\sqrt{2}}$$

$$\left| \frac{V_o}{V_{in}} \right| = 0.707$$

The gain of circuit is reduced to 0.707 times of maximum gain.



In General,

Roll-off rate of n th order filter is.

-n 20 db/dec.



7. Design a Butterworth with a cutoff frequency of 1 kHz.

Sol: $f_h = \frac{1}{2\pi RC} = 1 \times 10^3 \text{ Hz}$

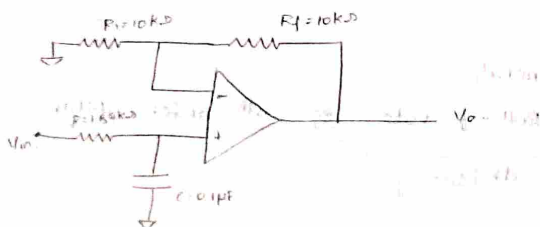
Let us take, $C = 0.1 \mu\text{F}$

$$10^3 = \frac{1}{2 \times 3.14 \times R \times 0.1 \times 10^{-6}}$$

$$R = \frac{1}{6.28 \times 10^{-7} \times 10^3}$$

$$R = 0.159 \times 10^4$$

$$R = 1.59 \text{ K}\Omega$$



here, $1 + \frac{R_f}{R_i} = A_f = 1$

assume $R_i = R_f = 10 \text{ K}\Omega$

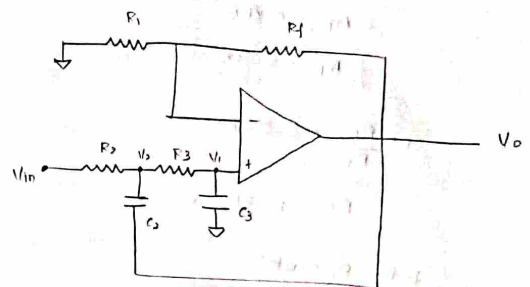
$$1 + \frac{R_f}{R_f} = A_f$$

$$1 + 1 = A_f$$

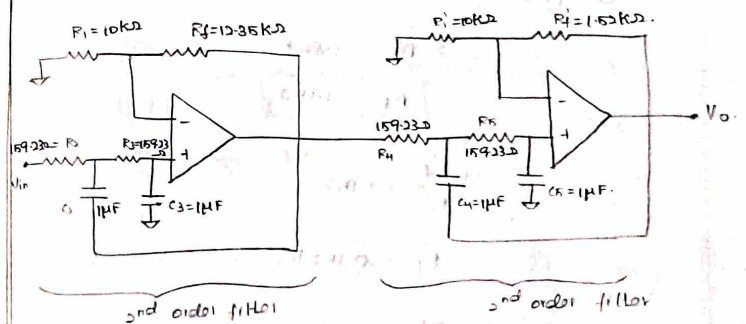
$$\text{gain, } A_f = 2$$

2nd order Butterworth LPF:

The order of Butterworth LPF which strongly no. of RC section connected in the circuit.



8. Design a Butterworth 4th order LPF with cutoff frequency of 1 kHz.



Sol:

Given, $f_h = 1 \text{ KHz}$

$$W.K.T, (s^2 + 0.765s + 1)(s^2 + 1.848s + 1)$$

↳ ①.

From 2nd order Butterworth LPF,

$$s^2 + \frac{(3-A_F)}{R_C} s + \frac{1}{R_C^2} \rightarrow \textcircled{2}$$

compare ① & ②, for 1st & 2nd order.

$$3 - A_F = 0.765$$

$$A_F = 2.235$$

$$1 + \frac{R_f}{R_1} = 2.235$$

$$R_f = 1.235 R_1$$

$$\text{let } R_1 = 10 \text{ k}\Omega$$

$$R_f = 12.35 \text{ k}\Omega$$

compare ① & ③, for 2nd (2nd order)

$$3 - A_F = 1.848$$

$$A_F = 1.152$$

$$1 + \frac{R_f'}{R_1'} = 1.152$$

$$R_f' = 0.152 R_1'$$

$$\text{let } R_1' = 10 \text{ k}\Omega$$

$$R_f' = 1.52 \text{ k}\Omega$$

$$\text{let us assume, } R_2 = R_3 = R_4 = R_5 = R$$

$$C_2 = C_3 = C_4 = C_5 = C$$

$$f_h = \frac{1}{2\pi RC} = 10^3$$

$$\text{let } C = 1 \mu\text{F}$$

$$R = \frac{1}{6.28 \times 10^3 \times 10^{-6}}$$

$$R = 0.159 \text{ k}\Omega$$

$$R = 159.23 \Omega$$

Difference btw 1st & 2nd order LPF:

1. RC sections.

For 1st order - 1 RC section

For 2nd order - 2 RC sections.

2. Roll-off rate.

1st order = -20 dB/dec.

2nd order = -40 dB/dec.

For nth order LPF:

$$\left| \frac{V_o}{V_{in}} \right| = \frac{1}{\sqrt{1 + (f/f_h)^{2n}}} ; f_h = \frac{1}{2\pi RC}$$

$$\left| \frac{V_o(s)}{V_i(s)} \right| = \left| \frac{V_o(j\omega)}{V_i(j\omega)} \right| = |H(j\omega)|$$

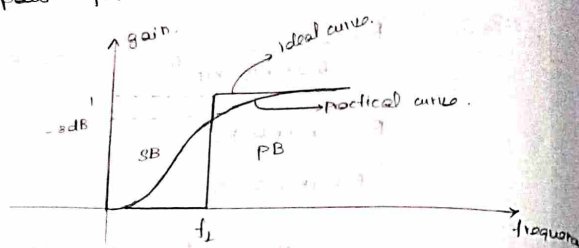
→ it allows low freq & it eliminates high

frequency (noise)

it is used in band-limited signal

HPF $\rightarrow$ It allows all the frequency above the cut-off frequency.

High pass filter:

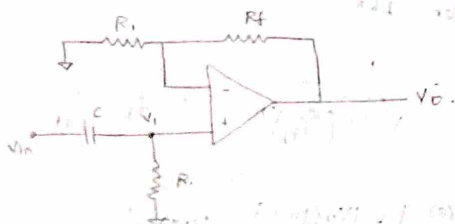


Exchange capacitor & Resistor in LPF

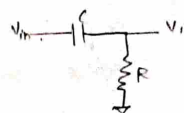
we get HPF

The filter which can allow all signal above the cutoff frequency and attenuate all the frequency signal below the cutoff frequency.

1st order active High pass filter.



$$V_o = \left(1 + \frac{R_f}{R_i}\right) V_i$$



$$V_i = \frac{V_{in}}{R + 1/sC} \cdot R$$

$$V_i = V_{in} \cdot \left(\frac{RSC}{1 + SRC} \right) \rightarrow (1)$$

$$V_o = \left(1 + \frac{R_f}{R_i}\right) V_i$$

$$= A_F \cdot \frac{V_{in}(SRC)}{1 + SRC}$$

$$\frac{V_o}{V_{in}} = A_F \cdot \frac{SRC}{1 + SRC}$$

$$\therefore S = j\omega$$

$$\omega = 2\pi f$$

$$\frac{V_o}{V_{in}} = A_F \cdot \left(\frac{j2\pi fRC}{1 + j2\pi fRC} \right)$$

$$\boxed{\frac{V_o}{V_{in}} = A_F \cdot \frac{j(f/f_L)}{1 + j(f/f_L)}}$$

$$\text{here, } f_L = \frac{1}{2\pi RC}$$

$f_L \rightarrow$ cut-off frequency of HPF.

case (i): $f \ll f_L$

$$\left| \frac{V_o(s)}{V_i(s)} \right| = \frac{A_c (f/f_L)}{\sqrt{1 + (f/f_L)^2}} \quad \text{[It's gain is very less]}$$

$$\left| \frac{V_o(s)}{V_i(s)} \right| < A_F \rightarrow \text{reduced.}$$

$\therefore$ It attenuates all the signal frequencies. $f \ll f_L$

case (ii): $f \gg f_1$

$$\left| \frac{V_o(s)}{V_i(s)} \right| > A_F$$

$$V_o(s) > A_F V_i(s)$$

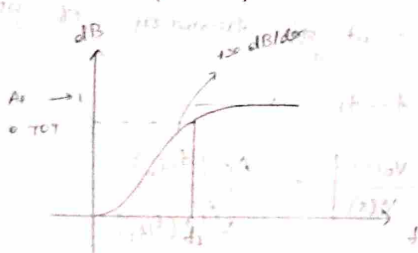
It allows all the signal frequency greater than cut-off frequency

case (iii): $f = f_1$

$$\left| \frac{V_o(s)}{V_i(s)} \right| = \frac{A_F (f/f_1)}{\sqrt{1 + (f/f_1)^2}}$$

$$= \frac{A_F (1)}{\sqrt{1+1}} = \frac{A_F}{\sqrt{2}}$$

$$\left| \frac{V_o(s)}{V_i(s)} \right| = 0.707 A_F$$



Design a 1st HPF the cut off frequency of 2 kHz.

$$f_1 = 2 \text{ kHz}$$

$$f_1 = 2 \times 10^3 \text{ Hz}$$

$$f_1 = \frac{1}{2\pi RC} = 2 \times 10^3$$

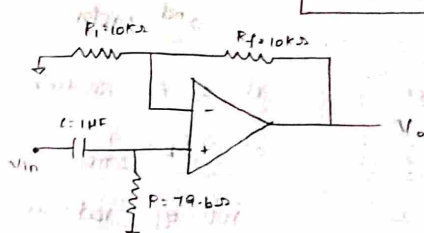
Let $C = 1 \mu\text{F}$,

$$R = \frac{1}{4 \times \pi \times 1 \times 10^{-6} \times 2 \times 10^3}$$

$$= \frac{1}{10.5663}$$

$$R = 0.094957 \text{ K}\Omega$$

$$R = 79.57 \Omega$$



$$A_F = 1 + \frac{R_F}{R_i} = 2$$

$$\frac{R_F}{R_i} = 1$$

$$R_F = R_i$$

Let us assume

$$R_i = R_F = 10 \text{ K}\Omega$$

Note:

* For n^{th} order high pass filter, the generalized expression of gain is

$$\left| \frac{V_o(s)}{V_i(s)} \right| = |H(s)| = \frac{A_F}{\sqrt{1 + (f_L/f)^{2n}}}$$

* For n^{th} order of LPF.

$$\left| \frac{V_o(s)}{V_i(s)} \right| = H(s) = \frac{A_F}{\sqrt{1 + (f/f_H)^{2n}}}$$

Difference b/w 1st & 2nd order HPF.

1st order

1. RC section

$$\Rightarrow f_L = \frac{1}{2\pi RC}$$

roll-off rate is

+20 db/dec.

2nd order

1. 2-RC section.

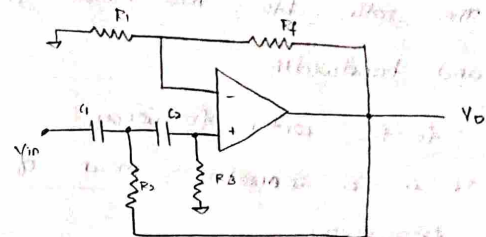
$$\Rightarrow f_H = \frac{1}{2\pi RC}$$

roll-off rate is

+40 db/dec.

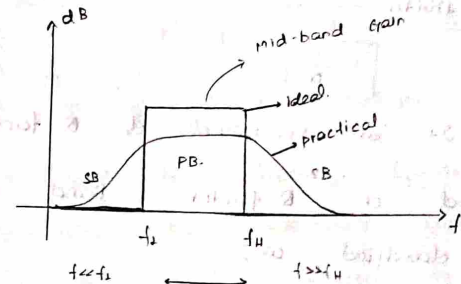
* For n^{th} order HPF. is +n20 db/dec.

2nd order high-pass filter:



Band-pass filter (BPF).

It allows all the signal frequency which is lies between two cut-off frequency. (f_L & f_H)



$\Rightarrow$ The signal is attenuated when $f < f_L$, $f > f_H$.

$\Rightarrow$ B/w f_L & f_H is called 'mid band frequencies'.

Bandwidth:

* The range of frequency for which (the gain is maximum) or (the circuit is providing maximum gain)

* It is used to analyze the performance of filter.

Quality factor:

* The ratio b/w mid-band frequency (f_0) and bandwidth.

$f_0 = f_c$ = center frequency.

$\Rightarrow$ It is a geometric mean of two cut-off frequencies.

$$Q = \frac{f_0}{BW} = \frac{f_c}{BW}$$

$$f_0 = f_c = \sqrt{f_L \cdot f_H}$$

selectivity:

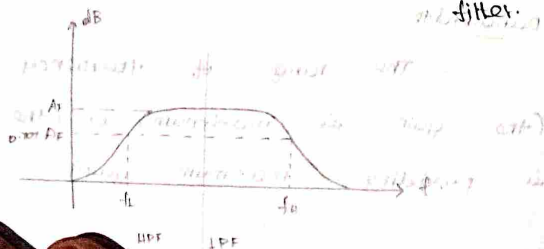
$$= \frac{1}{Q}$$

It is reciprocal of Q-factor.

• Based on Q-factor, Band pass filter is classified as,

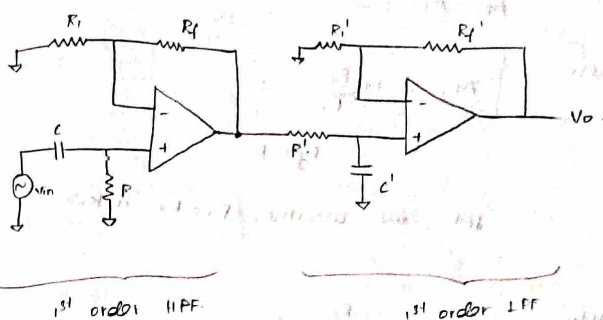
①. If $Q > 10 \rightarrow$ NBPF - Narrow band pass filter.

②. If $Q < 10 \rightarrow$ WBPF - wide band pass filter.



1st order wide Band Pass filter (WBPF).

* The HPF is cascaded with LPF.



$$\left| \frac{V_o(s)}{V_{in}(s)} \right| = |H(s)| = \frac{A_{F1} \left(\frac{f}{f_L} \right)}{\sqrt{1 + \left(\frac{f}{f_L} \right)^2}} \cdot \frac{A_{F2}}{\sqrt{1 + \left(\frac{f}{f_H} \right)^2}}$$

$\therefore$ gain of WBPF = Product of HPF & LPF

②. Design a wide BPF having $f_L = 400 \text{ Hz}$ and $f_H = 2 \text{ kHz}$ and pass band gain of 4. also find the quality factor of each filter circuit.

Sol:

$$f_L = 400 \text{ Hz}$$

$$f_H = 2 \text{ kHz}$$

WBPF gain of 4.

$$Q = ?$$

$$11 = A_{F1} - A_{F2}$$

$$\therefore A_{F1} = A_{F2} = 2$$

here, $A_{F1} = 1 + \frac{R_f}{R_i} = 2$

$$R_f = R_i$$

let us assume, $R_i = R_f = 10 \text{ k}\Omega$

also, $A_{F2} = 1 + \frac{R_f'}{R_i'} = 2$

$$\frac{R_f'}{R_i'} = 1$$

$$R_f' = R_i'$$

let us assume, $R_i' = R_f' = 10 \text{ k}\Omega$

for HPF, $f_c = 400 \text{ Hz}$

$$f_c = \frac{1}{2\pi RC} = 400$$

let $C = 1 \mu\text{F}$

$$R = \frac{1}{400 \times 2 \times \pi \times 10^{-6}}$$

$$= \frac{1}{5013.27} \times 10^6$$

$$= 3.9788 \times 10^{-4} \times 10^6$$

$$R = 397.88 \Omega$$

for LPF, $f_H = 5 \text{ kHz}$

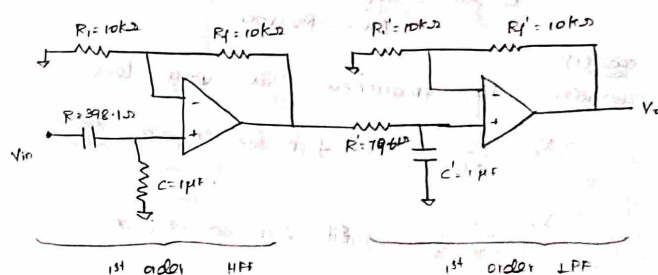
$$f_H = \frac{1}{2\pi R' C'} = 5 \times 10^3$$

let $C = 1 \mu\text{F}$

$$R' = \frac{1}{4 \times \pi \times 10^3 \times 1 \times 10^{-6}}$$

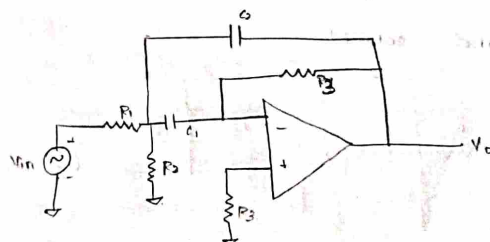
$$= 0.07961 \times 10^3$$

$$R' = 79.61 \Omega$$



Narrow Band pass filter: (NBPF)

⇒ The input is given in inverting terminal.



① op-amp is configured in inverting mode.

② It has two - feed back.

because of this a feedback it is named as multiple feedback filter.

③. C_1 is responsible for make this circuit fn as LPF.

C_2 is responsible for make this circuit fn as HPF.

(4). $Q > 10 = \frac{f_0}{\text{BW}} \leftarrow \text{Narrow}$

case (i):

consider if frequency is very less.

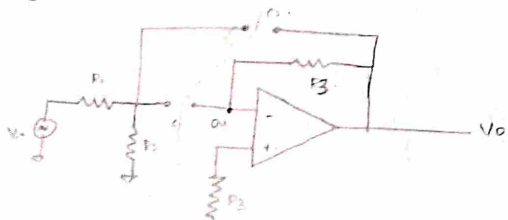
$$X_{C1} = \frac{1}{2\pi f C_1}, \text{ if } f=0 \Rightarrow X_{C1} = \infty.$$

$$X_{C2} = \frac{1}{2\pi f C_2}, \text{ if } f=0 \Rightarrow X_{C2} = \infty.$$

$\therefore$ If $X_{C1} = X_{C2} = \infty$ it act as open circuit.

$V_0 = 0$ due to virtual

ground concept.

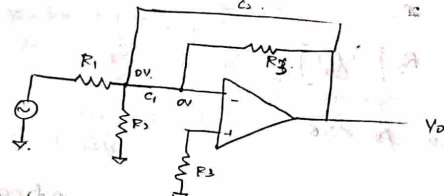


$\Rightarrow$ It exhibits the behaviour of HPF.

case (ii):

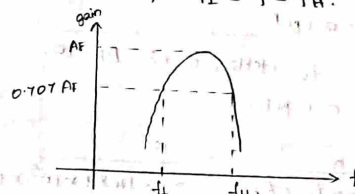
consider $f_{in} > f_H$

$X_{C1} \& X_{C2} = 0$ [it act as closed switch/circuit].
 $\Rightarrow$ It exhibits the behaviour of LPF.



case (iii):

consider, $f_L < f < f_H$.



advantage:

* By changing R_3 we can transient the center frequency of NBPF from one value to another value.

steps: Design parameters for NBPF:

1). assume $C_1 = C_2 = C$.

2). $R_1 = \frac{0}{2\pi f_0 C \cdot AF} = \frac{0}{2\pi f_c \cdot C \cdot AF}$

$$3). R_2 = \frac{Q}{2\pi f_0 C [2Q^2 - AF]}$$

$$4). R_3 = \frac{Q}{\pi f_0 C}$$

$$5). A_F = \frac{R_3}{2R_1}$$

$$6). R_3' = R_3 \left[\frac{f_0}{f_0'} \right]$$

also $AF < 2Q^2$.

$f_0 \rightarrow$ initial value of centre frequency
change to

$f_0' \rightarrow$ final value
change $R_3 \rightarrow R_3'$

11. Design a NBPB with centre frequency of 1 kHz, $Q = 13$ and $AF = 10$ also assume $C = 0.1 \mu F$.

Sol: Given, $f_0 = 1 \text{ kHz}$, $Q = 13$, $AF = 10$.

1). $C_1 = C_2 = 0.1 \mu F$.

$$2). R_1 = \frac{Q}{2\pi f_0 C AF} = \frac{13}{2 \times 3.14 \times 1 \times 10^3 \times 10^{-7} \times 10} = \frac{13}{6.28}$$

$$R_1 = 2.07 \text{ k}\Omega$$

$$3). R_2 = \frac{Q}{2\pi f_0 C [2Q^2 - AF]} = \frac{13}{6.28 \times 10^3 \times 10^{-7} [2 \times 169 - 10]} = \frac{13}{6.28 \times (328) \times 10^{-4}} = \frac{13}{2059.84} \times 10^4 = 6.3111 \times 10^{-4} \times 10^3$$

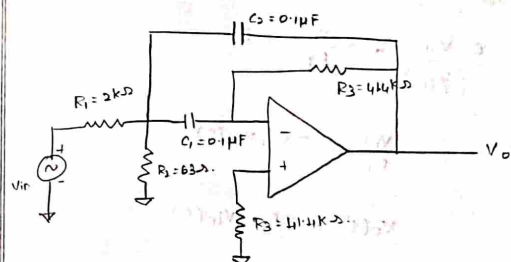
$$R_2 = 63.111 \Omega$$

$$4). R_3 = \frac{Q}{\pi f_0 C} = \frac{13}{3.14 \times 1 \times 10^3 \times 10^{-7}} = 4.1401 \times 10^4$$

$$R_3 = 41.401 \text{ k}\Omega$$

$$5). A_F = \frac{R_3}{2R_1} = \frac{41.401 \times 10^3}{2 \times 2.07 \times 10^3} = \frac{41.401}{4.14} = 10.000$$

$$AF = 10$$

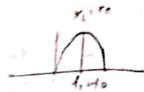


Find R_3' from $f_0 = 1 \text{ kHz}$ to 10 kHz .

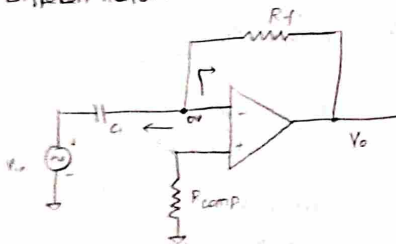
$$R_3' = R_3 \left[\frac{f_0}{f_0'} \right] = 63.111 \times \frac{1 \times 10^3}{10 \times 10^3}$$

$$R_3' = 6.311 \Omega$$

Resonance - at which the gain is max
($V_2 = x_c$)



Differentiator:



$\Rightarrow$ It differentiates the applied i/p.

Apply Nodal equation,

$$\frac{0 - V_{in}}{1/sC_i} + \frac{0 - V_o}{R_f} = 0$$

$$\frac{V_o(s)}{R_f} = -sC_i V_{in}(s)$$

$$V_o(s) = -sR_f C_i V_{in}(s)$$

$$\frac{V_o(s)}{V_i(s)} = -R_f C_i s$$

Inverse Laplace of V_o

$$\mathcal{L}^{-1}[V_o(s)] \Rightarrow V_o(t) = -R_f C_i \frac{dV_{in}(t)}{dt}$$

* Expression of ideal differentiator.

$$\text{Gain, } \frac{V_o(s)}{V_i(s)} = -R_f C_i (j\omega)$$

$$\left| \frac{V_o(s)}{V_i(s)} \right| = R_f C_i \omega$$

$$= R_f C_i 2\pi f$$

$$\left| \frac{V_o(s)}{V_i(s)} \right| = \frac{f}{f_0}$$

$$\text{where, } f_0 = \frac{1}{2\pi R_f C_i}$$

* Gain expression of ideal differentiation.

Limitation of ideal differentiator:

$\Rightarrow$ when (i) signal frequency $\uparrow$ so when (about)

(ii) the $\rightarrow$ i/p impedance.
 $\searrow$ gain.

* i/p frequency $\uparrow$ so - gain is more, it leads unstable.

* for high i/p frequency - i/p impedance is very low becoz of this the circuit will be affected by high frequency noise.

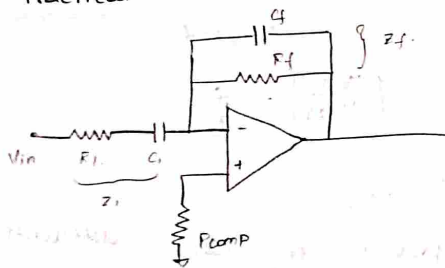
$\Rightarrow$ To overcome this we add

one Resistor, so that the minimum impedance is 'R' when high frequency.

... .. limit free noise.

⇒ to improve the stability we add one capacitor in feedback.

Practical differentiator:



$$Z_f = R_f \parallel \frac{1}{sC_f}, \quad Z_i = R_i + \frac{1}{sC_i}$$

$$\frac{V_o}{V_{in}} = -\frac{Z_f}{Z_i}$$

$$Z_i = \frac{sR_i C_i + 1}{sC_i}$$

$$\frac{1}{Z_f} = \frac{1}{R_f} + \frac{1}{1/sC_f}$$

$$= \frac{1}{R_f} + sC_f$$

$$= \frac{1 + sR_f C_f}{R_f}$$

$$Z_f = \frac{R_f}{1 + sR_f C_f}$$

$$\frac{V_o}{V_{in}} = -\frac{R_f}{1 + sR_f C_f} \cdot \frac{sC_i}{sR_i C_i + 1}$$

$$\frac{V_o}{V_{in}} = \frac{-sR_f C_i}{(1 + sR_f C_f)(1 + sR_i C_i)}$$

$$\left| \frac{V_o}{V_{in}} \right| = \frac{sR_f C_i}{(1 + sR_f C_f)(1 + sR_i C_i)}$$

assume, $R_f C_f = R_i C_i$

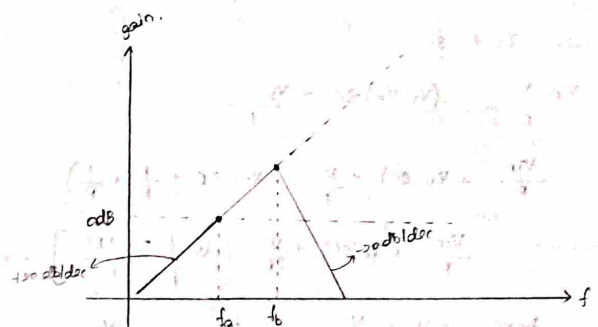
$$\left| \frac{V_o}{V_{in}} \right| = \frac{sR_f C_i}{[1 + sR_i C_i]^2}$$

$$s = j\omega = j2\pi f$$

$$= \frac{sR_f C_i}{[1 + j2\pi f R_i C_i]^2}$$

$$\left| \frac{V_o}{V_{in}} \right| = \frac{sR_f C_i}{[1 + j(f/f_b)]^2}$$

$$\text{here, } f_b = \frac{1}{2\pi R_i C_i} = \frac{1}{2\pi R_f C_f}$$



$$f = f_a, \quad A = 1$$

$$A = 0 \text{ dB}$$

conditions to design:

→ $f_b = 10 f_a$

→ signal frequency $f_s = \frac{f_a}{20}$

→ $T \geq R_f C_1$

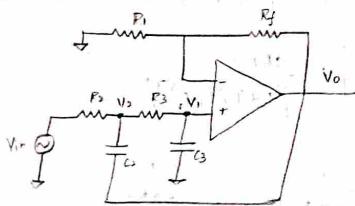
→ $R_1 C_1 = R_f C_f$

→ use f_a find R_f

→ use $f_b = 10 f_a$ find f_b

→ use f_b find R_1

2nd order low pass filter:



$$V_o = \left[1 + \frac{R_f}{R_1} \right] V_1 \quad \text{--- (1)}$$

let, $R_2 = R_3 = R$, $C_2 = C_3 = C$

$$I_1 = I_2 + I_3$$

$$\frac{V_{in} - V_2}{R} = (V_2 - V_o) sC + \frac{V_2 - V_1}{R}$$

$$\frac{V_{in}}{R} + V_o(sC) + \frac{V_1}{R} = V_2 \left(sC + \frac{1}{R} + \frac{1}{R} \right)$$

$$\frac{V_{in}}{R} + V_o(sC) + \frac{V_1}{R} = V_2 \left[\frac{2 + sRC}{R} \right] \quad \text{--- (2)}$$

$$\text{now, } V_1 = \frac{V_2}{R + 1/sC} \cdot \frac{1}{sC} = \frac{V_2}{1 + sRC}$$

$$\therefore V_2 = V_1 (1 + sRC)$$

apply this into equ (2)

$$\frac{V_{in}}{R} + V_o(sC) + \frac{V_1}{R} = \frac{V_1 (1 + sRC) (2 + sRC)}{R}$$

$$\frac{V_{in}}{R} + V_o(sC) = \frac{V_1 [2 + sRC + 2sRC + (sRC)^2] - V_1}{R}$$

$$\frac{V_{in}}{R} + V_o(sC) = \frac{V_1}{R} [2 + 3sRC + (sRC)^2 - 1]$$

$$\frac{V_{in}}{R} + V_o(sC) = \frac{V_1}{R} [1 + 2sRC + (sRC)^2]$$

$$V_{in} + V_o(sRC) = V_1 [1 + 2sRC + (sRC)^2]$$

$$\therefore V_1 = \frac{V_{in} + V_o(sRC)}{1 + 2sRC + (sRC)^2}$$

apply this into equ (1)

$$V_o = A_f \left[\frac{V_{in} + V_o(sRC)}{1 + 2sRC + (sRC)^2} \right]$$

$$V_o = A_f \left[\frac{V_{in}}{1 + 2sRC + (sRC)^2} \right] + A_f \left[\frac{V_o(sRC)}{1 + 2sRC + (sRC)^2} \right]$$

$$V_o \left[1 - \frac{A_f(sRC)}{1 + 2sRC + (sRC)^2} \right] = A_f \frac{V_{in}}{1 + 2sRC + (sRC)^2}$$

$$V_o \left[\frac{1 + 2sRC + (sRC)^2 - A_f(sRC)}{1 + 2sRC + (sRC)^2} \right] = A_f \frac{V_{in}}{1 + 2sRC + (sRC)^2}$$

$$\therefore \frac{V_o}{V_{in}} = \frac{A_f}{1 + sRC [2 - A_f] + (sRC)^2}$$

$$\begin{aligned}\frac{V_o}{V_{in}} &= \frac{AF}{1 + sRC[3 - AF] + s^2 R^2 C^2} \\ &= \frac{AF}{R^2 C^2 \left[s^2 + \frac{s}{RC} (3 - AF) + \frac{1}{R^2 C^2} \right]} \\ &= \frac{AF / (RC)^2}{s^2 + \left[\frac{3 - AF}{RC} \right] s + \frac{1}{R^2 C^2}} \\ &= \frac{AF \omega_h^2}{s^2 + \alpha \omega_h s + \omega_h^2} \quad \left(\begin{array}{l} \omega_h = \frac{1}{RC} \\ \alpha = 3 - AF \end{array} \right)\end{aligned}$$

$$\frac{V_o}{V_{in}} = AF \cdot \frac{1/R^2 C^2}{s^2 + \left[\frac{3 - AF}{RC} \right] s + \frac{1}{R^2 C^2}} \rightarrow (3)$$

we know that for 2nd order system.

$$\frac{V_o}{V_{in}} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{s^2 + \sqrt{2}s + 1} \rightarrow (4)$$

compare (3) & (4).

$$\frac{3 - AF}{RC} = \sqrt{2}$$

Let $RC = 1$, $3 - AF = \sqrt{2}$.

$$AF = 3 - \sqrt{2}$$

$$AF = 1.586$$

For nth order LPT, $\left| \frac{H(j\omega)}{A_0} \right| = \frac{1}{\sqrt{1 + (\omega/\omega_h)^{2n}}}$

Also, $\omega_h^2 = \omega_0^2 = \frac{1}{R^2 C^2}$

$$\omega_0 = \frac{1}{RC}$$

$$f_0 = \frac{1}{2\pi RC}$$

If $R_1 = R_2 = R$, $C_1 = C_2 = C$

$$f_0 = \frac{1}{2\pi \sqrt{R_1 R_2 C_1 C_2}}$$

design a second order butterworth lowpass filter having upper cut-off frequency of 1 kHz.

Sol:

$$AF = 1 + \frac{R_f}{R_1} = 1.586$$

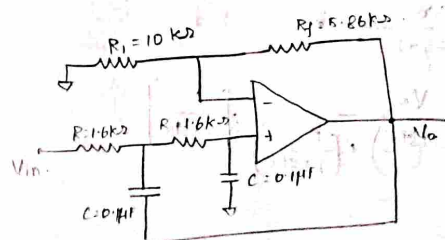
$$R_f = 0.586 R_1$$

Let $R_1 = 10 \text{ k}\Omega$, $R_f = 5.86 \text{ k}\Omega$.

$$f_0 = \frac{1}{2\pi RC} = 1000$$

Let $C = 0.1 \mu\text{F}$, $R = \frac{1}{2\pi \times 0.1 \times 10^{-6} \times 1000}$

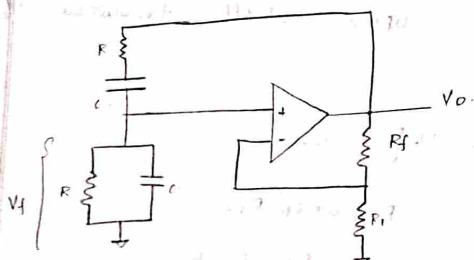
$$R = 1.6 \text{ k}\Omega$$



order factors of polynomials.

1. $s+1$
2. $s^2 + \sqrt{2}s + 1$
3. $(s+1)(s^2+s+1)$
4. $(s^2+0.765s+1)(s^2+1.848s+1)$
5. $(s+1)(s^2+0.618s+1)(s^2+1.618s+1)$

Wien-bridge oscillator.



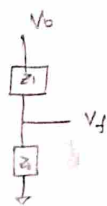
$$\text{Let } Z_1 = R + \frac{1}{sC} = \frac{1+sRC}{sC}$$

$$Z_2 = R \parallel \frac{1}{sC} = \frac{R \times 1/sC}{R + 1/sC} = \frac{R}{sRC + 1}$$

here, V_f = Voltage across ' Z_2 '

$$V_f = \frac{V_o}{Z_1 + Z_2} \cdot Z_2$$

$$= \frac{V_o}{\left(\frac{1+sRC}{sC}\right) + \left(\frac{R}{1+sRC}\right)} \cdot \left[\frac{R}{1+sRC}\right]$$



$$= \frac{V_o (1+sRC)(sC)}{\left[\left(\frac{1+sRC}{sC}\right) + \frac{R}{(1+sRC)}\right]} \cdot \frac{R}{(1+sRC)}$$

$$V_f = \frac{V_o (sRC)}{1 + (sRC)^2 + 2sRC + sRC}$$

$$V_f = \frac{V_o (sRC)}{1 + 3sRC + (sRC)^2}$$

$$\therefore \beta = \frac{V_f}{V_o} = \frac{sRC}{1 + 3sRC + (sRC)^2}$$

$$= \frac{sRC}{sRC \left[3 + sRC + \frac{1}{sRC}\right]}$$

$$= \frac{1}{3 + j\omega RC + \frac{1}{j\omega RC}}$$

$$\beta = \frac{1}{3 + j\left[\omega RC - \frac{1}{\omega RC}\right]}$$

to have sustained oscillations ' β ' must be a Real.

$$\beta = \frac{V_f}{V_o} = \frac{1}{3}$$

equating imaginary part to '0'.

$$\omega RC - \frac{1}{\omega RC} = 0$$

$$\omega^2 R^2 C^2 = 1$$

$$\omega^2 = \frac{1}{R^2 C^2} \Rightarrow \omega = \frac{1}{RC}$$

$$f_0 = \frac{1}{2\pi RC}$$

To satisfy the magnitude condition,

$$A_v \beta = 1.$$

$$A_v \left(\frac{1}{3}\right) = 1.$$

$$A_v = 3.$$

(3) Design a Wien bridge oscillator for

$$f_0 = 1 \text{ kHz}.$$

Sol: Given, $f_0 = 1 \text{ kHz} = 1000 \text{ Hz}.$

W.K.T $f_0 = \frac{1}{2\pi RC} = 1000$

Let $C = 0.1 \mu\text{F}.$

$$R = \frac{1}{2\pi \times 1000 \times 0.1 \times 10^{-6}}$$

$$R = 1.59 \text{ k}\Omega.$$

$$R \approx 1.6 \text{ k}\Omega.$$

for Wien-Bridge

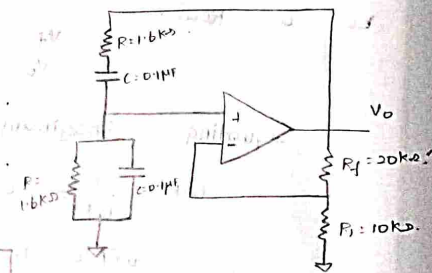
$$A_v = 3.$$

$$\left[1 + \frac{R_f}{R_1}\right] = 3.$$

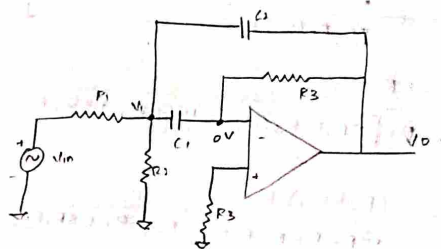
$$R_f = 2R_1.$$

Let $R_1 = 10 \text{ k}\Omega.$

$$\therefore R_f = 20 \text{ k}\Omega.$$



Narrow Band Pass Filter (NBPF).



Let, $Z_1 = 1/sC_1$, $Z_2 = 1/sC_2$

write Nodal equation w.r.t Node V_1

$$\frac{V_1 - V_{in}}{R_1} + \frac{V_1}{R_2} + (V_1 - V_o) sC_2 + V_1 (sC_1) = 0.$$

$$V_1 \left[\frac{1}{R_1} + \frac{1}{R_2} + sC_2 + sC_1 \right] - \frac{V_{in}}{R_1} - V_o (sC_2) = 0 \rightarrow (1)$$

write Nodal eqn w.r.t Node V_o

$$(0 - V_1) sC_1 + \frac{(0 - V_o)}{R_3} = 0.$$

$$\frac{V_o}{R_3} = -V_1 sC_1$$

$$V_1 = \frac{-V_o}{sR_3C_1} \rightarrow (2)$$

apply equation (2) into equation (1)

$$\left[\frac{-V_o}{sR_3C_1} \right] \left[\frac{1}{R_1} + \frac{1}{R_2} + sC_2 + sC_1 \right] - \frac{V_{in}}{R_1} - V_o (sC_2) = 0.$$

$$V_o (sC_2) + \frac{V_o}{sR_3C_1} \left[\frac{1}{R_1} + \frac{1}{R_2} + sC_2 + sC_1 \right] = -\frac{V_{in}}{R_1}$$

$$V_o \left[sC_2 + \frac{1}{sR_3C_1} \left(\frac{R_2 + R_1 + sR_1R_2C_2 + sR_1R_2C_1}{R_1R_2} \right) \right] = -\frac{V_{in}}{R_1}$$

$$V_o \left[\frac{s^2 R_1 R_2 R_3 C_1 C_2 + R_2 + R_1 + s R_2 R_1 C_2 + s R_1 R_2 C_1}{s R_1 R_2 R_3 C_1} \right] = -\frac{V_{in}}{R_1}$$

$$\frac{V_o}{V_{in}} = \frac{-s R_1 R_2 R_3 C_1}{R_1 [s^2 R_1 R_2 R_3 C_1 C_2 + R_2 + R_1 + s R_2 R_1 C_2 + s R_1 R_2 C_1]}$$

$$\frac{V_o}{V_{in}} = \frac{-(R_2 R_3 C_1) s}{s^2 R_2 R_3 R_1 C_1 C_2 + R_2 + R_1 + s R_2 R_1 C_2 + s R_1 R_2 C_1}$$

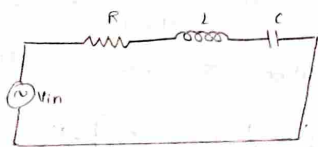
$$\frac{V_o}{V_{in}} = \frac{-(R_2 R_3 C_1) s}{R_1 R_2 R_3 C_1 C_2 \left[s^2 + \frac{1}{R_1 R_3 C_1 C_2} + \frac{1}{R_2 R_3 C_1 C_2} + \frac{s}{R_3 C_1} + \frac{s}{R_2 C_2} \right]}$$

Let, $C_1 = C_2 = C$

$$\frac{V_o}{V_{in}} = \frac{-(s/R_1 C_0)}{s^2 + \frac{1}{R_1 R_3 C^2} + \frac{1}{R_2 R_3 C^2} + \frac{2s}{R_3 C}}$$

$$\therefore \frac{V_o}{V_{in}} = \frac{(-s/R_1 C)}{s^2 + \frac{2s}{R_3 C} + \frac{1}{R_3 C^2} \left[\frac{1}{R_2} + \frac{1}{R_1} \right]} \rightarrow (3)$$

Transfer fn of RLC series resonant circuit.



Overall impedance.

$$Z = R + sL + \frac{1}{sC} = \frac{sRC + s^2 LC + 1}{sC}$$

$$V_o = \frac{V_{in} \cdot AF}{\left[\frac{sRC + s^2 LC + 1}{sC} \right]}$$

$AF \rightarrow$ Max Gain at resonance

$$= \frac{V_{in} (sRC) \cdot AF}{LC \left[s^2 + \frac{R}{L} s + \frac{1}{LC} \right]}$$

Note:
 $\Rightarrow$ at resonance Max
o/p occurs
across the resistor
'R'.

$$V_o = \frac{V_{in} (R/L) s \cdot AF}{LC \left[s^2 + (R/L) s + 1/LC \right]}$$

$$\therefore \frac{V_o}{V_{in}} = \frac{-s (R/L) \cdot AF}{s^2 + \left(\frac{R}{L} \right) s + \frac{1}{LC}} \rightarrow (4)$$

$\therefore V_o = V_R$

here, Quality factor, $Q = \frac{\omega_0 L}{R}$

$$\therefore \frac{R}{L} = \frac{\omega_0}{Q} \rightarrow (5)$$

$$\text{Also } \omega_0^2 = \frac{1}{LC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \rightarrow (6)$$

apply equ (5), (6) into equ (4).

$$\therefore \frac{V_o}{V_{in}} = \frac{-s (\omega_0/Q) \cdot AF}{s^2 + \left(\frac{\omega_0}{Q} \right) s + \omega_0^2} \rightarrow (7)$$

The above equation, (7) represents the transfer fn of RLC series resonant circuit.

Compare equ (7) with equ (3).

$$AF \cdot \frac{\omega_0}{Q} = \frac{1}{R_1 C}$$

$$\therefore R_1 = \frac{Q}{\omega_0 C \cdot AF} = \frac{Q}{2\pi f_0 C \cdot AF} \rightarrow (8)$$

$$\text{Hence } \frac{\omega_0}{Q} = \frac{2}{R_3 C}$$

$$R_3 = \frac{2Q}{\omega_0 C} = \frac{2Q}{2\pi f_0 C} = \frac{Q}{\pi f_0 C}$$

$$R_3 = \frac{Q}{\pi f_0 C} \rightarrow (9)$$

$$\text{Also, } \omega_0^2 = \frac{1}{R_3 C^2} \left[\frac{1}{R_2} + \frac{1}{R_1} \right]$$

$$\omega_0^2 [R_3 C^2] = \frac{1}{R_2} + \frac{1}{R_1}$$

$$\frac{1}{R_2} = \omega_0^2 [R_3 C^2] - \frac{1}{R_1}$$

$$\frac{1}{R_2} = \omega_0^2 [R_3 C^2] - \frac{1}{(Q/2\pi f_0 C A_F)}$$

$$= \omega_0^2 R_3 C^2 - \frac{2\pi f_0 C A_F}{Q}$$

$$\frac{1}{R_2} = \frac{Q\omega_0^2 [R_3 C^2] - 2\pi f_0 C A_F}{Q}$$

$$\therefore R_2 = \frac{Q}{Q\omega_0^2 (R_3 C^2) - 2\pi f_0 C A_F}$$

$$= \frac{Q}{Q\omega_0^2 \left[\frac{Q}{\pi f_0 C} \right] C^2 - 2\pi f_0 C A_F}$$

$$= \frac{Q}{Q\omega_0^2 \left[\frac{2Q C^2}{2\pi f_0 C} \right] - 2\pi f_0 C A_F}$$

$$= \frac{Q}{2Q^2 \omega_0 C - 2\pi f_0 C A_F} \quad \because \omega_0 = 2\pi f_0$$

$$= \frac{Q}{\omega_0 C [2Q^2 - A_F]}$$

$$\therefore R_2 = \frac{Q}{2\pi f_0 C [2Q^2 - A_F]} \rightarrow (10)$$

from equ (9),

$$f_0 = \frac{Q}{\pi R_3 C} \rightarrow (11)$$

Hence from equ (2),

$$f_0 = \frac{Q}{2\pi R_1 C A_F} \rightarrow (12)$$

equating (11) & (12),

$$\frac{Q}{\pi R_3 C} = \frac{Q}{2\pi R_1 C A_F}$$

$$\therefore A_F = \frac{R_3}{2R_1} \rightarrow (13)$$

to change centre frequency from $f_0 \rightarrow f_0'$

$$\therefore R_2' = R_2 \left[\frac{f_0}{f_0'} \right]^2$$